

QUANTIFYING LOTTERY CHOICE COMPLEXITY*

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Abstract

We develop interpretable, quantitative indices of the objective and subjective complexity of lottery choice problems. These indices capture the predicted error rate in identifying the lottery with the highest expected value. The most important complexity feature is the state-by-state dissimilarity of the lotteries in the set (“tradeoff complexity”). Using our complexity indices, we study the behavioral signatures of complexity out-of-sample across one million decisions in 11,000 unique binary choice problems. Complexity strongly predicts cognitive uncertainty, choice inconsistencies and attenuation of decisions to problem fundamentals. This attenuation can produce systematic biases in revealed preference measures such as spurious risk aversion. These effects are large, to the degree that complexity explains a larger fraction of estimated choice errors than proximity to indifference. Accounting for complexity-linked attenuation in structural estimations improves model fit substantially. Complexity aversion explains a smaller fraction of the data.

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1 Introduction

Much recent research in both the lab and the field emphasizes that decision problems involving risk are often complex, and that this complexity matters for choice. This raises the question: which risky choice problems actually are complex, and how does complexity affect decisions? The typical approach in the literature is to proceed on a case-by-case basis: the researcher intuits a specific complexity feature and investigates how it affects behavior. Yet, ideally, one would like to quantify the overall complexity of a choice set, and to study the predictive power of such a composite notion of complexity.

In this paper, we develop an empirical mapping from choice set features to indices of choice complexity, which can be computed for any binary lottery choice dataset. We use these indices to study choice behavior and document that complexity-linked behavioral attenuation – an insensitivity of decisions to fundamentals – is quantitatively even more important for predicting choice than popular behavioral models such as prospect theory.

Development of complexity indices. The literature has not yet converged on a widely-accepted definition of “complexity.” Contributions outside economics often characterize complexity as the difficulty of aggregating—or predicting the aggregate behavior of—many interrelated constituent components (e.g. Weaver, 1948). Applied to lottery choice, this notion would correspond to the difficulty of aggregating state probabilities and payoffs into a choice. However, this difficulty is latent and unobserved. For the purposes of this paper, we thus adopt the convention that “complex” simply means that the problem has features that make it likely for people to make mistakes, regardless of what the nature of those mistakes is. This includes different types of noise, stable heuristics, or shying away from options one cannot properly evaluate.

In choice data, mistakes are generally unobserved: the researcher does not know the decision-maker’s utility function and, as a result, cannot distinguish errors from preferences. To circumvent this problem, we propose to quantify the objective complexity of a lottery choice set as *the predicted error rate in identifying the lottery with*

the *highest expected value*, where the prediction is computed as a function of choice set features.

This index does not rest on the assumption that people are necessarily risk neutral. We also don't assume that maximizing expected value (EV) is as complex as maximizing expected utility. Rather, the thought is that those choice set features that make it harder to gauge EV also make it harder to gauge expected utility (or other subjective values) because the cognitive process of aggregating probabilities and payouts is similar across the two tasks. Based on this idea, we formally state an identifying assumption for the true complexity of standard lottery choice problems to be a monotone function of our complexity index.

Quantifying complexity based on an EV task has two advantages. First, relative to revealed-preferences measures such as random choice, it does not impose a specific behavioral model of the nature of people's mistakes. For instance, the EV task captures mistakes that result from both noise and stable heuristics. Second, relative to subjective difficulty measures, our approach does not require awareness of mistakes and is incentivized.

To train our index, we implement experimental problems in which subjects' payout only depends on whether they successfully identify the lottery with the highest EV in the set. Because ex ante we do not know which choice set features drive complexity, we design a large-scale experiment in which subjects make decisions involving 2,220 quasi-randomly generated problems.

In selecting the features that enter our objective complexity index (*OCI*), we balance the tradeoff between completeness and interpretability that is inherent to any predictive index. We implement exploratory LASSO regressions to identify the most predictive features, and then construct a handcrafted index using a sparse set of easily interpretable features that closely approximates the performance of machine learning algorithms.

OCI comprises two classes of features: (i) the proximity of the EVs of the lotteries in the set; and (ii) features that intuitively capture the difficulty of aggregating (and trading off) the constituent components of the problem. In this latter class, the most important feature by far is the *excess dissimilarity* of the lotteries in a set, by which

we mean the degree to which the cumulative distribution functions of two lotteries are dissimilar from each other above and beyond their difference in expected value. This measure can intuitively be understood as capturing the strength of tradeoffs across different payout states (“tradeoff complexity”). At the extreme, lotteries with a first-order stochastic dominance relationship have excess dissimilarity of zero because they do not involve any tradeoffs. Excess dissimilarity fundamentally captures a *comparative* evaluation process.

In binary choice sets that comprise a safe option, excess dissimilarity essentially captures the variance of the risky option. This implies that – if one option is safe – variance is associated with complexity.

In addition to *objective* choice complexity, we also quantify *subjective* complexity. To do this, we elicit subjects’ cognitive uncertainty (*CU*) in the EV task and use this data to develop a subjective complexity index (*SCI*). In principle, *OCI* and *SCI* capture distinct concepts, but in practice they are very similar. Both complexity indices are primarily intended as *predictive tools* to predict and interpret choice, rather than as causal identification devices.

To illustrate how our indices can be deployed to assess and compare the degree of complexity across papers, we calculate the complexity of ten lottery choice experiments from the literature. We see large variation in complexity, both across papers and across treatments within the same paper.

Evidence for construct validity. The key identifying assumption behind our index – that mistake rates in a choice problem are increasing functions of mistake rates in the analogous EV problem – is difficult to test directly. However, we use a variety of exercises to provide tentative evidence for a link between the EV task and choice problems. We implement standard binary lottery choice experiments and elicit subjects’ *CU* about their choices (Enke and Graeber, 2023). When the same set of subjects completes the same problems in both the choice design and the EV design, the correlation between average problem-level *CU* across the two tasks is $r = 0.72$. We also show that *OCI* and *SCI* are strongly predictive of variation in *CU* out-of-sample (in a different pool of subjects). Finally, we find that the effect of individual complexity features such as excess dissimilarity on *CU* is similar in lottery choice and

the EV task, which again tentatively suggests a link between the cognitive difficulty of the two problem types.

Behavioral attenuation. We study the predictive power of our complexity indices out-of-sample in traditional binary lottery choice problems. We both collect our own dataset and re-analyze the most comprehensive dataset on binary lottery choice in the literature (Peterson et al., 2021). In total, we evaluate one million decisions across 11,000 unique choice problems.

The most important complexity signature is *behavioral attenuation*: an insensitivity of choice to problem fundamentals. This attenuation effect is large: the elasticity of choice to the difference in EVs of the two lotteries (or the difference in estimated prospect-theory values) decreases by about 75% going from low to high complexity.

Behavioral attenuation could reflect that complexity is linked to either choice noise or stable heuristics. To investigate this, we link our indices to rich data on the frequency of within-subject choice inconsistencies in repeated elicitations of the same problem. We find that our index is strongly correlated with this proxy for choice noise ($r \approx 0.5$). Indeed, *OCI* explains an order of a magnitude more of the across-problem variation in proxies for choice errors (such as choice inconsistency) than the estimated proximity to indifference in a prospect theory model.

Spurious risk aversion and risk love. How do people respond to the complexity of individual lotteries? Ex ante, there are two plausible possibilities: stronger attenuation and systematically disliking complex options (complexity aversion). We find evidence for the importance of both. First, like previous work, we find pronounced complexity aversion to compound lotteries (e.g., Halevy, 2007; Gillen et al., 2019). The number of payout states, on the other hand, is largely uncorrelated with choice, reminiscent of the mixed evidence in the literature summarized by Wakker (2022).

Second, we find pronounced evidence for variance-linked behavioral attenuation: choice is substantially more responsive to EV differences (or estimated CPT value differences) when the variance of the lottery is smaller. We document that this can produce spurious small-stakes risk aversion or risk love. Intuitively, because higher lottery variance makes choice noisier, it pushes choice rates towards 50%, which can

generate either spurious risk aversion (if the lottery is very attractive) or spurious risk love (if the lottery is unattractive). As a result, as in other recent work, complexity-driven attenuation (or noise) does not “cancel out” but produces systematic bias (e.g., Andersson et al., 2016; Gillen et al., 2019).

Structural estimations. To further assess quantitative magnitudes, we conduct structural estimations. To study complexity-linked attenuation, we allow the responsiveness parameter in a logit model to be a function of complexity, which amounts to introducing one additional parameter. This generates an increase in model fit of 32%. In our dataset, this increase is even slightly larger than the combined increase resulting from all of prospect theory (reference dependence, value function curvature, loss aversion and probability weighting). Intuitively, the reason why models that ignore complexity exhibit considerably lower performance is that they underpredict the probability that people choose the (estimated) higher value option when complexity is low, but overpredict it when complexity is high.

On the other hand, allowing for complexity aversion increases model fit by 3%.

Contribution and relation to prior work. A number of theoretical contributions have proposed that lottery complexity depends on problem features such as entropy or support (e.g., Mononen, 2021; Hu, 2023; Puri, 2025; Gabaix and Graeber, 2023).¹ Our complexity metrics differ substantially from theirs, both because we construct composite measures and because our indices include excess dissimilarity as a main feature.

The broad idea that dissimilarity produces mistakes originates in perceptual psychology experiments, such as when subjects estimate which of two rectangles is larger (Tversky and Russo, 1969). Natenzon (2019) and He and Natenzon (2022) develop multi-attribute choice models in which choice noise increases in dissimilarity. Fishburn (1978) and Shubatt and Yang (2024) present lottery choice models that feature dissimilarity. In psychology, a small collection of papers have shown that dissimilarity

¹Appendix Table 4 provides an overview of the experimental literature on complexity in lottery choice. This literature has shown that different lottery features can produce either aversion, or the adoption of stable rules, or higher noisiness. Our comprehensive approach that relies on multiple complexity features allows us to highlight that there is no such thing as a “single complexity response”. This said, we do find that behavioral attenuation is quantitatively by far the most important pattern.

is linked to estimated choice noise (Buschena and Zilberman, 2000; Erev et al., 2002, 2010).

Our contributions to these literatures are twofold. First, we construct the first comprehensive indices of the objective and subjective complexity of lottery choice problems. We do this (i) by identifying complexity features based on an objective process rather than based on researcher intuitions and (ii) by quantifying the relative importance of different features. While the lottery choice literature has often focused on support as a measure of complexity, we find that excess dissimilarity is quantitatively more important for predicting errors. Because our indices are transparent and defined on objective choice set features, they can be computed in a standardized fashion across datasets. We make available code that automates this process. In Section 7, we discuss potential applications of the complexity indices.

Our second contribution is to study behavioral signatures of complexity in a dataset that is orders of magnitude larger and more comprehensive than typical experimental datasets. This second contribution involves (iii) showing that complexity overwhelmingly predicts attenuation, with some evidence also for complexity aversion; (iv) documenting that attenuation produces systematic biases in estimated risk aversion; and (v) quantifying the importance of complexity for model fit through structural analyses.

Our results are related to recent contributions that have documented links between an attenuation of decisions and complexity and / or cognitive uncertainty (e.g., Enke and Graeber, 2023; Enke et al., 2025, 2024; Frydman and Jin, 2021; Vieider, 2021, 2022; Yang, 2023; Bastianello and Imas, 2025). Here, we offer an index that predicts the magnitude of attenuation, choice inconsistencies and cognitive uncertainty based on objective problem features.

Section 2 lays out a conceptual framework. Section 3 describes the data and Section 4 develops the indices. Section 5 discusses behavioral signatures of complexity and Section 6 presents structural estimations. Section 7 concludes.

2 Conceptual Framework

2.1 Approach and Identifying Assumption

Consider choice sets comprising two lotteries, A and B , and denote by $d_c \in \{A, B\}$ the decision maker’s (DM’s) choice. We denote by $U(x)$ the DM’s *true* utility of a lottery, where we allow utility to include any form of non-standard preferences (e.g., cumulative prospect theory). We are interested in quantifying the complexity of choosing between A and B . Rather than exogenously impose what counts as complex, we wish to measure which choice set features contribute to complexity. We simply define the objective complexity (OC) of choosing from a menu involving lotteries A and B as the probability that the DM makes a mistake, and we assume that this probability depends on features of the menu.

$$OC_{A,B} = P(d_c \notin \underset{x \in \{A,B\}}{\operatorname{argmax}} U(x)) \quad (1)$$

Because the researcher usually does not know the DM’s objective function, $OC_{A,B}$ cannot be directly observed. To overcome this problem, we consider a second, ancillary decision problem in which the DM is tasked with identifying which lottery has the highest EV. We denote the decision by d_s . In this deterministic task, the objective function is known, thus we can directly observe *deterministic* objective complexity (DOC).

$$DOC_{A,B} = P(d_s \notin \underset{x \in \{A,B\}}{\operatorname{argmax}} EV(x)) \quad (2)$$

This complexity metric has two features. First, it imposes no assumptions on the nature of the mistakes. As summarized in Appendix Table 4, prior research has emphasized at least two structurally different types of phenomena: noisy evaluations and stable heuristics. Because DOC is only defined on errors, we can capture both of these possible mechanisms. For example, a tendency to follow a heuristic such as “select lottery with highest minimum payout (max-min)” would generate predictable

errors in the EV task.² The fact that the EV task does not impose a specific behavioral model of a complexity response is an attractive feature relative to alternative conceivable approaches. For example, defining complexity based on observed choice inconsistencies would impose the assumption that complexity only triggers noise.

Identifying assumption. A main idea behind this paper is that choice errors arise in large part due to the latent difficulty of aggregating probabilities and payouts into a decision. Complexity arising from aggregation similarly arises in lottery choice and EV problems because the aggregated value of a lottery is not transparently provided to real decision makers. Instead, both in the EV task and in choice problems, the decision maker needs to integrate different payouts and associated probabilities to arrive at a choice. Our identifying assumption is that the frequency of errors in the EV task is monotonically related to errors in the choice task,

$$OC_{A,B} = f(DOC_{A,B}) , \quad (3)$$

with $f(\cdot)$ a monotone increasing function. This identification assumption does not require that people are necessarily risk neutral. It also doesn't require that maximizing EV is as difficult as maximizing expected utility. Instead, our assumption is that those choice set features that make it *more* difficult to gauge EV also make it *more* difficult to gauge expected utility, producing a link between error rates in the two different problems.

Our identifying assumption is a statement about how the *relative* difficulty of different choice problems is correlated with the relative difficulty of different EV problems, rather than about how the absolute difficulty of a choice problem compares to the absolute difficulty of the analogous EV problem. In particular, our identifying assumption is consistent with situations in which (i) choosing from the set $C_1 = \{A, B\}$ is hard and (ii) determining the highest EV option in C_1 is easy, as long as, for the set $C_2 = \{D, E\}$, a higher difficulty of choosing in C_2 compared to C_1 goes hand-in-hand with a higher difficulty of identifying the highest EV option in C_2 compared to C_1 .

²We note that by focusing on mistakes, we will not code a problem as complex if decision-makers use stable heuristics that consistently lead to the *correct* answer.

2.2 Empirical Implementation

Because it is impractical for researchers to always implement an analogous EV problem to quantify the complexity of their choice problem of interest, we leverage the idea that error rates in the EV problem can be *predicted* based on choice set features. This is attractive because once complexity is defined based on objective features of the set, it can be easily computed for any standard dataset.

For a choice set involving two lotteries, A and B , denote by $f^{A,B}$ an N -dimensional vector of choice set features, where the first element is a constant. Denote by $\epsilon_{A,B}$ a mean-zero disturbance term.

Definition 1. *For a sample of EV menus with element $\{A,B\}$, use OLS to estimate $\hat{\beta}$ and $\hat{\gamma}$ in*

$$DOC_{A,B} = \sum_{i=1}^N \beta_i f_i^{A,B} + \gamma \ln(1 + |EV(A) - EV(B)|) + \epsilon_{A,B}.^3 \quad (4)$$

Then, the objective complexity index for choice set $\{C,D\}$ is given by

$$OCI_{C,D} := \sum_{i=1}^N \hat{\beta}_i f_i^{C,D} + \hat{\gamma} \ln(1 + |EV(C) - EV(D)|). \quad (5)$$

This index has a simple interpretation once it is applied to standard lottery choice problems: it captures the error rate in the analogous EV problem that is predicted by the choice set features. We allow the absolute EV difference to enter non-linearly because in random choice models such as logit, mistake rates are a concave function of the absolute EV difference.⁴

OCI comprises both features that capture the proximity of the aggregated (expected) values, and features that capture the complexity of aggregating payouts and probabilities conditional on the EV difference. OCI does not allow the effect of the features f_i to depend on the EV distance. This is clearly a simplification. We expect our index to perform well in problems for which the DM is neither extremely close

³Formally, the estimated $\hat{\beta}$ and $\hat{\gamma}$ are specific to the sample of EV menus on which they're estimated. When we estimate our index in practice, we find very similar coefficients across different subsamples of interest.

⁴The $\ln(1+x)$ specification is inspired by Weber's law but otherwise arbitrary.

nor very far from indifference. Below in Section 6, we develop an alternative complexity index that is independent of the proximity of the aggregated values but requires stronger structural assumptions on decision errors.

All of the above concerns the quantification of *objective* choice complexity. Yet in many economically-relevant situations, it is *subjective* complexity that matters for behavior. We define a subjective complexity index ($SCI_{C,D}$) analogously to Definition 1, except that in equation (4) objective mistake rates are replaced by the subjective probability of making a mistake in the EV problem.

Tentative tests of identifying assumption. Because $OC_{C,D}$ is unobserved, we cannot directly test our identifying assumption. We here describe three exercises we undertake below to provide suggestive evidence that (objective and subjective) mistake rates in the choice and EV tasks are tightly linked.

First, we conduct a *within-sample* exercise. Here, we correlate subjective mistake probabilities (cognitive uncertainty) in a choice task with cognitive uncertainty of the same subjects in the analogous EV task. This offers some evidence on the identifying assumption in eq. (3), albeit only for subjective complexity.

Second, in an *out-of-sample* exercise, we correlate behavior in a choice task with the complexity indices that are trained on different subjects in a different sample of problems. Specifically, we correlate the objective complexity index $OCI_{C,D}$ (that is, the predicted error rate in the EV task) with observed choice inconsistencies in repetitions of the same choice task. Under the assumption that these choice inconsistencies are mistakes and proxy for $OC_{C,D}$, this approximates the identifying assumption for our objective index. Third, and similarly, we correlate the subjective complexity index $SCI_{C,D}$ (that is, the predicted subjective error rate in the EV task) with cognitive uncertainty in the choice task, which approximates the identifying assumption for our subjective index.

3 Experimental Datasets

Our main analysis is based on four experimental datasets, three of which we collected ourselves. Table 1 provides an overview.

Table 1: Overview of experiments and data sources

Experiment	Description	Problems	Subjects	Decisions
<i>EV Tasks</i>	EV problems & <i>CU</i> elicitation	2,100 procedurally generated & 120 targeted	1,148	57k
<i>Choice Tasks from PEA</i>	Lottery choice problems	10,398 procedurally generated	15,151	973k
<i>Choice Tasks</i>	Lottery choice problems & <i>CU</i> elicitation	500 procedurally generated	250	12.5k
<i>Within Subject</i>	EV and choice problems & <i>CU</i> elicitation	240 procedurally generated	300	12k

Notes. *PEA* = Peterson et al. (2021). *CU* = Cognitive uncertainty.

3.1 Experiment *EV Tasks*

Decision task. We present experimental participants with two or more lotteries. Instead of asking them to choose the lottery that they personally prefer, we instruct participants to indicate the lottery that has the highest EV. Each subject completed 50 decision problems. This design has been used previously (e.g., Benjamin et al., 2013), albeit always on a very small set of distinct problems. The task is similar in spirit to the “deterministic mirrors” proposed by Oprea (2024) and Martínez-Marquina et al. (2019).

We avoid jargon and never speak of “expected value.” Rather, we instruct participants to select the lottery that has the highest average payout if each lottery is run many, many times (100,000 times). We explain that, in each run, we record the payout of the lottery and then compute the average payout across runs. Subjects’ potential bonus equals \$10 if they select the correct lottery, and nothing otherwise. This incentive scheme has two main upsides. First, it makes transparent the objective nature of the task. Second, it holds the incentives constant across problems.

We believe the cognitive difficulty of lottery choice problems mainly stems from the difficulty of aggregation: whether deliberately or not, people ought to somehow combine the probabilities and payouts of different states to reach a decision. An attractive feature of our design is that this aspect of the decision problem is similar between our *EV Task* and real lottery choice problems.⁵

⁵At the end of the study, we asked subjects whether they had used a calculator or other help.

A potential concern with our design is that participants may misunderstand it and, instead, treat it as a standard choice task. We took the following measures to ensure that this was not the case. First, we deliberately designed the incentive scheme described above to make it clear that no lottery was ever actually being played out. We verified subjects’ understanding that their bonus depends on the accuracy of their decision rather than on the outcome of a lottery draw using two comprehension checks.⁶ Second, the question on subjects’ decision screen reads: “Which lottery has the highest average payout if the computer runs it many, many times?”, rather than “Which lottery do you select?”. Third, our instructions emphasized that the task has a mathematically correct solution. Fourth, if it was the case that some subjects had still misunderstood our instructions, we would expect them to exhibit risk aversion. We find no evidence for this in our data.

Cognitive uncertainty elicitation. On each decision screen, we additionally asked “How certain are you that each lottery has the highest average payout?”, and subjects distributed 100 “certainty points” across the lotteries in the set to indicate their probabilistic beliefs. Our instructions clarified that subjects should express how likely they think it is that each lottery has the highest average payout. This procedure is similar to the elicitation of cognitive uncertainty (*CU*) in Enke and Graeber (2023) and Enke et al. (2023, 2024, 2025).⁷

Generation of problems. We desire our complexity indices to be applicable across different datasets. It is, hence, crucial for us to develop them on a dataset that includes as many commonly-encountered lottery features as possible. We designed the experiment *EV Tasks* to comprise a total of 2,220 unique problems. A first set

23% of subjects indicated they had. We have verified that our complexity indices are very similar if we restrict attention to the sub-sample who report not having used external help.

⁶Subjects who did not get all of our four comprehension check questions right on the second try were immediately routed out of the experiment (8% of prospective participants were routed out because they failed twice, and 14% of our final sample failed their first attempt).

⁷There are lingering concerns in the literature that subjects may misunderstand complex incentive schemes. To account for this, we incentivize *CU* in two different ways. In 15% of the sample, we deployed a binarized scoring rule with a prize of \$10 and a winning probability of $q = 1 - (1 - g)^2$, where g is the probability assigned to the correct option. In 85% of the sample, we instead paid subjects \$0.10 for each point they allocated to the correct lottery. This scoring rule is not proper but simple to understand. The distributions of *CU* and response times in these two sub-samples are virtually identical, as are mistake rates. Appendix Figure 8 shows a screenshot of a decision screen.

of 2,100 problems was generated using a quasi-random procedure, meaning that the lotteries are random conditional on a set of parameters that we impose to (i) make the problems non-trivial and (ii) ensure variation across a large set of features. This random procedure is called for because we as researchers do not know *ex ante* which features matter most for complexity, and because we do not want our own intuitions to constrain the development of the indices.

The remaining choice problems were devised by following the typical approach in lab experiments of designing a relatively small number of problems that are targeted at identifying some specific effect of interest. Our main analyses will leverage all choice problems in the dataset. To conserve space, we relegate a separate analysis of the smaller, targeted set to Appendix H of our earlier NBER working paper (#31677).

We focus on the 2,118 two-item menus and discuss an extension to larger menus in Section 7. In the two-item menus, 58% involve deciding between a safe payment and a lottery. In the remaining 42% of problems, both lotteries are non-degenerate. In our sample, the number of states in a lottery can vary between two and seven. We collected data until each problem was completed by at least 20 subjects (median is 22 and average 26).

Summary statistics. The average problem-level error rate in the *EV Tasks* experiment is 27%, with a median of 25% and $IQR = [14\%, 38\%]$. The average problem-level subjective error rate (average *CU*) is 18%, with a median of 17% and $IQR = [13\%, 22\%]$. The correlation between problem-level error rates and average *CU* is $r = 0.49$ ($p < 0.01$), suggesting subjects’ beliefs are reasonably well-calibrated, on average. In only 7.9% of two-item menus does a majority select the wrong option. This majority is statistically significant in only 1.4% of problems. Appendix Table 8 presents further summary statistics.

3.2 Lottery Choice Problems

Dataset of Peterson et al. (2021). Peterson et al. used a quasi-random procedure to generate the 10,398 unique binary choice sets that we use.⁸ 15,151 Amazon

⁸We drop problems that involve ambiguity or that involve two safe payments. We combine identical problems with / without feedback.

Mechanical Turk (AMT) workers completed an average of 13 problems five times each, for an average total of 65 decisions per subject. The dataset was designed to span a much larger space of choice problems than previous data-collection exercises, making the data well-suited for our purposes. Appendix Table 8 presents summary statistics.

While the richness and size of this dataset provide many advantages, it has the downside that Peterson et al. (2021) did not pay out losses and instead truncated all payouts from below at zero. While this is a shortcoming, we view it as ultimately inconsequential: (i) the results shown below are robust to restricting attention to choice problems that only involve gains and (ii) the results are very similar in our own incentivized experiments described next. Appendix C.3 replicates our main results excluding choice problems from PEA that involve losses.

Experiment Choice Tasks. We generated 500 choice problems using a similar automated quasi-random procedure as in the *EV Tasks* experiment, except that we only implemented binary choice sets. Losses were deducted from a budget.

In addition to asking subjects to choose between the two lotteries, we also elicited their *CU*, asking how certain they are (in percentage terms) that they selected the option they actually prefer (Enke and Graeber, 2023). Appendix Figure 10 shows a screenshot. Each subject completed 50 choice problems.

Experiment Within Subject. Each subject encountered 20 problems in both as a choice problem and as an EV problem. Subjects first completed all 20 choice problems or all 20 EV problems, where the order of choice and EV task was randomized across subjects.⁹ We generated 240 distinct choice sets, each of which was completed by 25 subjects, on average. To study robustness to payout procedures, in the EV task part of this experiment, subjects were paid out the EV of the lottery they selected, rather than receiving \$10 if they made the correct decision.

⁹*CU* was measured using the following questions. EV task: “How certain are you that you actually selected the lottery with the highest average payout?” Choice task: “How certain are you that you actually prefer the lottery you chose above?”

3.3 Implementation

Our own experiments were conducted on Prolific. See Appendix G for screenshots of instructions and comprehension check quizzes. In *EV Tasks*, subjects earned a completion fee of \$6, and 1 in 2 subjects was randomly selected to be eligible for a bonus of \$10 if they made the correct choice on a (uniformly) randomly selected decision. In *Choice Tasks* and *Within Subject*, participants received a fixed payment of \$3.50. In addition, 1 in 5 subjects were randomly selected to be eligible for a bonus wherein we randomly selected one decision to be payout-relevant. We pre-registered the predictions and sample size for experiments *Choice Tasks* and *Within Subject* on aspredicted.org under #130662 and #173455. We didn't pre-register the *EV Tasks* experiment because there was no specific hypothesis: we use these data to create complexity indices, rather than to show that a specific feature would matter.

4 Development of the Complexity Indices

4.1 Choice Set Complexity

We train the complexity indices on a randomly selected subset of 75% of all EV problems (train set) and use the remaining 25% as a test set. A main question is which features should be included in the indices. To balance the typical tradeoff between interpretability and completeness, we proceed in three steps.

1. *Exploratory LASSO index*: We assemble a large vector of features. Because many of these features will be intra-correlated (giving rise to multicollinearity), we estimate LASSO regressions, such that only a relatively small number of features will have non-zero coefficients.
2. *Handcrafted index*: We inspect the LASSO-generated index and approximate it based on only a handful of simple and easily interpretable features.
3. *Assess completeness*: Following Fudenberg et al. (2022), we benchmark the handcrafted index against a machine learning ensemble.

Exploratory LASSO indices. Appendix B.2 provides a list of all 44 choice set features that we consider. When a feature is defined over a single lottery (such as a lottery’s variance), we compute the average value in the set.¹⁰ We also consider non-linear transformations of these averages (log and square).¹¹ We only consider primitive features of the lotteries, i.e. payouts and probabilities. We don’t allow interactions between features (as we will see below, they do not matter much).¹²

Some of the features we consider may affect choice for standard expected utility reasons, such as a lottery’s variance. However, as emphasized in Section 5.3, the way in which features like variance predict choice through complexity is distinct from what expected utility theory prescribes. Moreover, because our indices are developed based on the *EV Task*, utility curvature cannot drive any feature’s inclusion in an index.

We train the LASSO using a 10-fold cross-validation procedure, choosing the penalty parameter to minimize mean cross-validated error in the training set. Appendix Table 6 reports the results of the LASSO regressions. Because many of the features are highly intra-correlated (e.g., range and variance of payouts), the particular features that get selected by the LASSO should not be viewed as uniquely important, but instead as representative of broad classes of important features. We do not assign causal interpretations to any complexity features – given that many features are almost mechanically related, it is not straightforward to imagine what it would mean to marginally change one feature while holding all others constant. Rather, our objective is to craft a composite index that *predicts* well.

Handcrafted indices. This observation motivates us to develop handcrafted, interpretable versions of the complexity indices that are based on fewer features, each of which represents a broad class of features that we now discuss. While we recognize that manually selecting features raises potential concerns over artificially generating “desired” results, we view these as ultimately inconsequential because the handcrafted indices turn out to be almost perfectly correlated with the exploratory LASSO indices.

¹⁰Computing averages appears justified because mistake rates increase in the variance of both option A and B, where B is always the option with a smaller number of payout states.

¹¹Whenever we say we compute the log of x , we mean that we compute $\ln(1 + x)$.

¹²We also don’t allow features of choice sets encountered in the past. This appears justified because, compared to the efficient coding literature (e.g., Frydman and Jin, 2021), we (i) implement many fewer trials and (ii) vary many more features, both of which make it less likely that subjects differentially learn higher cognitive precision for some problems than others.

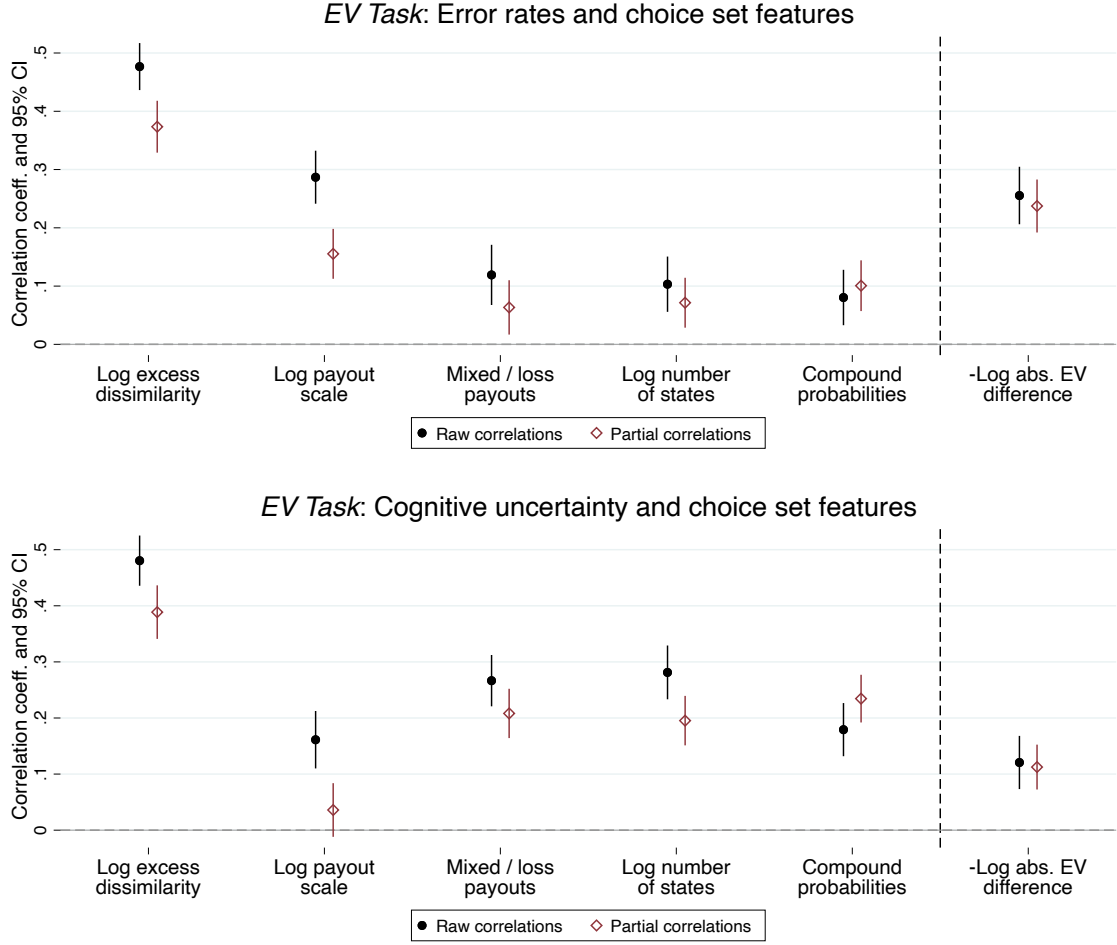


Figure 1: Raw and partial correlation coefficients between task-level error rates / average *CU* and choice set features in the train set in the *EV Tasks* experiment (1,587 unique problems). Whiskers show 95% confidence intervals. Partial correlations are calculated controlling for all of the other features in the figure. Log scale, mixed / loss payouts and log number of states are computed as averages across the lotteries in a set.

Appendix Table 7 reports OLS estimates of eq. (4) using our hand-selected features.¹³ Our indices *OCI* and *SCI* correspond to the fitted values of these regressions. To present the results in a more intuitive way, Figure 1 reports correlation coefficients between error rates (or *CU*) and those choice set features that enter our final indices. Appendix Figure 17 reports the analogous results for the *Within Subjects* experiment.

¹³Our procedure implicitly assumes that a single index captures complexity across all different types of menus. In ancillary analyses, we experimented with a finite mixture approach in which different types of menus are assigned different complexity indices. The results show that the resulting “within-type” indices are highly correlated ($r \geq 0.95$) with *OCI*. Our indices are also virtually identical when we derive them on the sub-sample of problems in which one option is a safe payment, or on the sub-sample in which both lotteries are non-degenerate.

Excess dissimilarity: The strength of tradeoffs across states. The stand-out predictor of both objective and subjective complexity is the excess dissimilarity between the lotteries in a set, by which we mean the degree to which lotteries are dissimilar from each other above and beyond their difference in expected value. As illustrated in the top left panel of Figure 2, the so-called “Wasserstein 1-distance” between the CDFs of two lotteries is given by $\delta_{A,B} = \int_{\mathbb{R}} |F_A(x) - F_B(x)| dx$. We then define excess dissimilarity as: $d_{A,B} = \delta_{A,B} - |EV(A) - EV(B)|$. To illustrate, the following lottery pairs have low excess dissimilarity. Option A: “Get \$20 with probability 80%”, and Option B: “Get \$21 with probability 70%”. In contrast, Option B’: “Get \$70 with probability 21%” has high excess dissimilarity with A, even though B and B’ have the same EV. Thus, choosing between A and B is predicted to be simpler than choosing between A and B’.

Excess dissimilarity is large when the lotteries have very different advantages and disadvantages. We find it helpful to think of this measure as capturing “tradeoff complexity” – complexity as it arises from the difficulty of aggregating tradeoffs across different payout states. Excess dissimilarity equals zero when there are no tradeoffs across states (i.e., when differences in EV arise due to first-order stochastic dominance). We can loosely think of excess dissimilarity as a measure of how “close” lotteries A and B are to having a dominance relationship. Excess dissimilarity is fundamentally an object that captures the difficulty of *comparative* evaluations: how the two lotteries perform in their worst state, their best state, their “median” state, and so on.¹⁴

Researchers often think of “similar” as “difficult”. The key distinction is that here “similarity” does not refer to the proximity to indifference (similarity of *aggregated* values) but, instead, to the similarity of the *disaggregated* objects, netting out the similarity in aggregate value.

Figure 1 shows the raw correlation between log excess dissimilarity and error rates as well as average CU , which is approximately $r = 0.48$ in both cases. Log excess dissimilarity is the strongest predictor of both errors and CU in our data and thus

¹⁴We are implicitly putting the two lotteries into a common, perfectly correlated state space. Formally, we think of the state x as a draw from a Uniform distribution on $[0, 1]$, and we say the lotteries A and B are “perfectly correlated” in that they return $F_A^{-1}(x)$ and $F_B^{-1}(x)$, respectively.

the most important component of our complexity indices. The correlation between dominance and error rates is $r = 0.27$, meaning that the more general dissimilarity measure almost doubles explanatory power.

The top right and bottom panels of Figure 2 illustrate how variation in excess dissimilarity – as induced by variation in payouts and probabilities – predicts mistakes in the EV task. Each plot shows a binned scatter plot that controls for the absolute EV difference between the two options in the set (because excess dissimilarity is dissimilarity net of absolute expected values difference).

In the top right panel, we consider the decision between a safe payment and a binary lottery, where (to illuminate the role of payouts) the probability of the upside is restricted to be in $[0.4, 0.6]$. The plot shows that as the spread between the lottery upside and downside increases (which increases the dissimilarity between the lottery and the safe payment), mistake rates strongly increase. This insight will be important below because the spread of payouts (or lottery variance) is essential in the estimation of risk aversion.

In the bottom left panel, we illustrate the role of payout probabilities. Again, we consider the decision between a safe payment and a binary lottery, and vary the probability of the lottery upside. Dissimilarity between a lottery and a safe payment is small when the probabilities are very extreme. Consistent with the idea that low dissimilarity produces fewer mistakes, we see that mistake rates are strongly hump-shaped, with intermediate probabilities around 50% producing the highest error rates.

Finally, to illustrate that it is not generically true that intermediate probabilities contribute to dissimilarity and complexity, we consider the decision between two lotteries, A and B, where the probability of B’s upside is in $[0.4, 0.6]$. The bottom right panel shows mistake rates as a function of the probability of A’s upside. In this case, dissimilarity is lowest when A’s upside probability is intermediate. Indeed, mistake rates are now U-shaped. Viewed in combination, the bottom left and bottom right panels illustrate that complexity is fundamentally based on what a lottery is compared with.

Payout scale. Much research on number perception suggests that people find it harder to work with larger numbers (Weber’s law). This intuitively adds to the

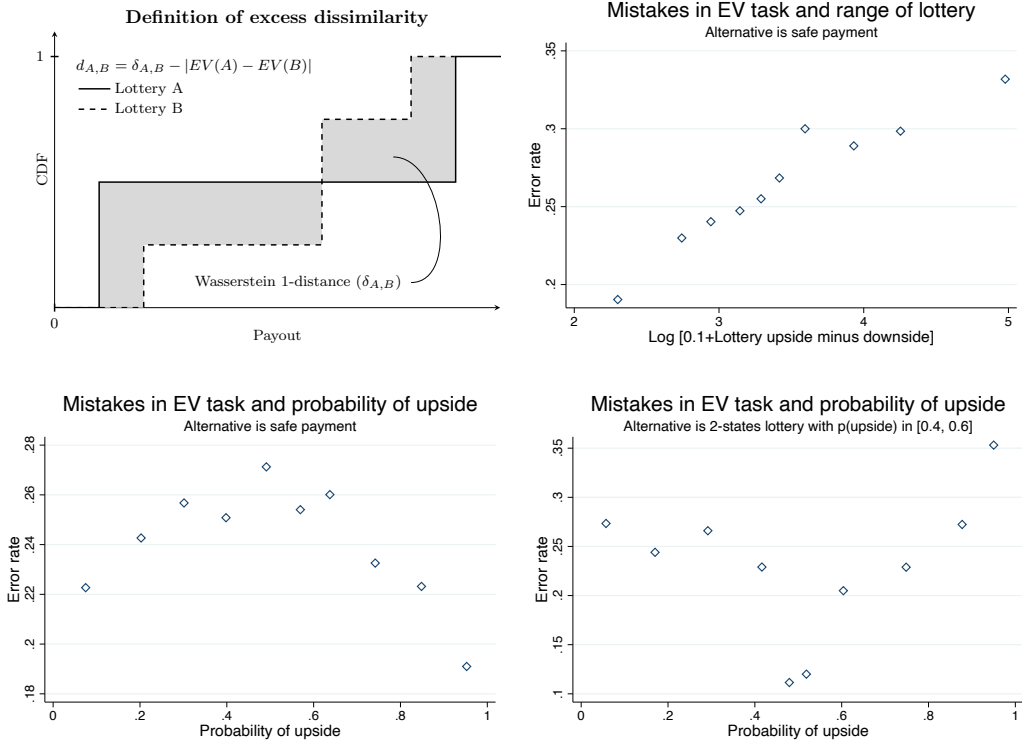


Figure 2: Top left panel: Definition of excess dissimilarity. Top right panel: Binned scatter plot of error rate in EV task as a function of difference between lottery upside and downside, when decision is between a binary lottery (with probability of upside in $[0.4, 0.6]$) and a safe payment. Bottom left panel: Binned scatter plot of error rate in EV task as a function of probability of lottery upside, when decision is between binary lottery and safe payment. Bottom right panel: Binned scatter plot of error rate in EV task as a function of probability of lottery upside, when decision is between binary lottery and another binary lottery with probability of upside in $[0.4, 0.6]$. All binned scatter plots control for the absolute EV difference.

difficulty of integrating different payouts and probabilities. Our preferred measure of payout scale for an individual lottery is the log average absolute payout; for the choice-set measure, we average this individual measure across the set.

Mixed and loss gambles. It appears cognitively harder for people to process negative payouts. In our data, both pure loss gambles and mixed gambles produce significantly higher error rates and CU , compared to pure gains menus. To keep our indices sparse, we generate one variable to capture these patterns, which is the fraction of lotteries in the set that includes at least one negative payout.

Support. Figure 1 shows that the average (log) number of states in the choice set is significantly correlated with error rates and CU . We include this variable in our complexity indices, but we note that – while a main focus of the literature – it is a

considerably less important determinant of overall complexity than some of the other features.

Compound probabilities. The presence of compound lotteries intuitively leads to higher aggregation complexity because they require an additional computational step (reduction). We find that compound probabilities are associated with both higher error rates and higher CU .¹⁵

Proximity of expected values. Unlike the aforementioned features, the difference in EV between the two options *does* affect the magnitude of errors in standard random choice models. Figure 1 shows that proximity to indifference is indeed meaningfully correlated with errors and CU . Because this relationship is concave, we work with the natural log in the construction of our handcrafted indices. However, the link between proximity and errors (or CU) is relatively small compared to some of the features that capture aggregation complexity. This is a first indication of what we repeatedly emphasize throughout this paper: errors (and our complexity indices) largely reflect aggregation complexity rather than proximity to indifference.

Composite complexity indices. OCI and SCI consist of linear combinations of the features listed above, with weights given by the OLS coefficients in Appendix Table 7.¹⁶ In the *EV Tasks* test set data, OCI and SCI exhibit a raw correlation of $r = 0.87$, ($p < 0.01$). An important question is how complete the complexity indices are. In principle, they could be incomplete for two reasons. First, the list of features that the LASSO was based on may be incomplete. Second, interactions of features may play an important role. To assess completeness, we follow Fudenberg et al. (2022) and benchmark the out-of-sample performance of our complexity indices against that of a machine learning ensemble, including a black-box convolutional neural net. Appendix B discusses the details. In short, we find that OCI explains 95% of the variation relative to a machine learning benchmark, and that excess dissimilarity alone is 65% complete.

¹⁵For compound probability choice problems, one option involves an unknown probability p , which subjects know is drawn uniformly from some specified range.

¹⁶We winsorize all complexity indices from below at zero and from above at 0.5 because they capture predicted mistake rates.

4.2 Lottery-Specific Complexity

Our main index quantifies the complexity of a choice problem rather than of a single lottery. We believe this is important because – by the logic of dissimilarity causing complexity – the complexity of an individual lottery will depend on what it is compared against. Still, one useful special case that attracts much attention in theoretical and empirical practice is to quantify the complexity of a single lottery when the alternative is a safe payment. In particular, a natural benchmark is to compute the complexity of a lottery as the difficulty of choosing between the lottery and a safe payment that is just below (or above) its own EV.

We construct indices of objective and subjective lottery complexity (*OLCI* and *SLCI*) in an analogous fashion to *OCI* and *SCI*, except that they are computed based on a two-item ‘choice set’ that comprises a lottery and its own EV. Thus the indices of lottery complexity comprise as features a lottery’s support, scale, negative payouts, compound probabilities, and excess dissimilarity between a lottery and its own EV. Given that the dissimilarity between a lottery and its own EV basically amounts to a lottery’s variance, this means that – when the alternative is a safe payment – variance is a key predictor of a lottery’s complexity. Indeed, in our *EV Task*, the correlation between log lottery variance and error rates (when the alternative is a safe payment) is $r = 0.42$ ($p < 0.01$).

4.3 Tentative Evidence on Identifying Assumption

Within-sample evidence. In our *Within Subject* experiment, the same subject completes identical choice and EV problems. In a first step, we calculate average *CU* for each choice and EV problem and then correlate the two. The advantage of this is that averaging across subjects reduces measurement error. The raw correlation of average *CU* in choice and EV is $r = 0.72$ ($p < 0.01$). In a second step, we calculate the same correlation at the decision level, within each subject. We correlate each individual’s *CU* across the two problem formats, controlling for subject fixed effects. The partial correlation is $r = 0.30$ ($p < 0.01$), see Appendix Figure 16.¹⁷

¹⁷This correlation is still $r = 0.30$ when we control for (i) which lottery the subject chose in the choice task and (ii) which lottery they selected in the EV task. It’s also $r = 0.30$ when we restrict

That this decision-level correlation is substantially smaller than the problem-level correlation is perhaps unsurprising because the degree of measurement error in CU and resulting attenuation bias is likely much smaller in the problem-level analysis, where each observation is itself computed as an average from 25 subjects.

Next, we check whether each complexity feature (rather than the composite index) is linked to CU in choice and in the EV task in similar ways. Appendix Table 11 compares the correlations of the features in our indices with CU in choice and CU in the EV task. The results show that the correlations between CU and features are usually very similar. Excess dissimilarity, support and the presence of losses are all correlated with CU in choice and in the EV task in almost identical ways. Compound probabilities and payout scale, on the other hand, are associated with CU in the EV task but not in choice.

Out-of-sample evidence. Next, we check whether our complexity indices predict CU out-of-sample: in a different set of subjects and in a different set of problems than the indices were trained on. In our *Choice Tasks* data, average CU exhibits a correlation of $r = 0.63$ ($p < 0.01$) with SCI and of $r = 0.47$ ($p < 0.01$) with OCI , see Appendix Figure 11. In sum, the complexity indices are systematically linked to higher subjective error rates. Below in Section 5.2, we further link the complexity indices to a proxy for objective error rates (choice inconsistencies).

4.4 Comparing Task Complexity Across Experiments

One benefit of our index is that it can be deployed to compare the complexity of different experimental designs. We demonstrate by collecting designs from ten lottery choice papers in the literature. We make no claims of representativeness here, but instead heuristically selected these papers either because they are widely cited or because they study complexity. Nine of these designs use binary choice tasks (where we can compute OCI) and one uses multiple price list valuation tasks (where we instead compute $OLCI$). In Figure 3, we report the average task complexity in each of these designs. Given that OCI captures the predicted error rate in the analogous

attention to observations in which the subject chose a safe payment.

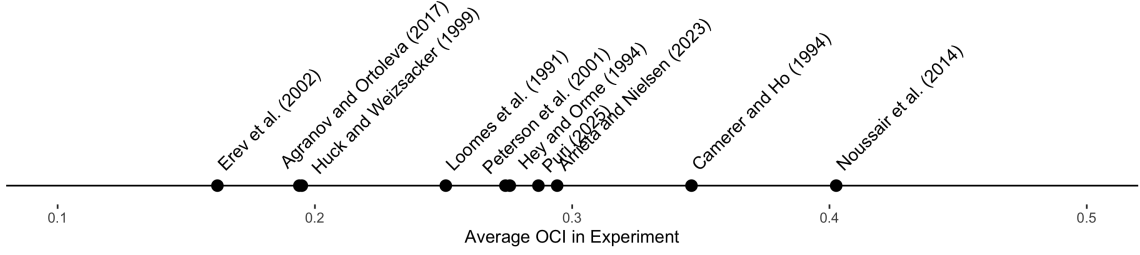


Figure 3: Average OCI across tasks for selected experimental designs from the literature. For Puri (2025), which uses multiple price list valuation rather than binary choice tasks, average OLCI is reported.

EV problem, the variation appears to be large. For instance, the range of *OCI* across studies is even larger than the interquartile range in the very large and comprehensive dataset collected by Peterson et al. (2021).

We also compare the average *OCI* for different treatment arms within individual papers. For instance, in Agranov and Ortoleva (2017), tasks in the *EASY* treatment have an average *OCI* of 0.04 vs. 0.36 in *HARD*; in Arrieta and Nielsen (2023), tasks in the *Simple2* treatment have an average *OCI* of 0.28 vs. 0.31 in *Complex*; and in Puri (2025), 2-state lotteries have an average *OLCI* of 0.275 vs 0.31 for 16-state lotteries. These results illustrate that if our complexity indices were standardly computed in lottery choice experiments going forward, they would provide a standardized metric along which papers can be compared.

5 Behavioral Signatures of Complexity

We next deploy the complexity indices to understand choice behavior. Unless noted otherwise, we pool the data from our own *Choice Tasks* experiment with those collected by Peterson et al. (2021). In our analyses, the level of observation is not an individual decision but, instead, choice rates in a unique choice problem.

We study the link between complexity and choice in three steps. First, we provide a few illustrative examples of low- and high-complexity problems and associated choice patterns. Second, we systematically study the predictive power of the index of objective complexity, *OCI*. Given the similarity between *OCI* and *SCI*, the results using *SCI* are essentially identical. In a third step, we study separate lottery features.

Table 2: Example choice problems

#	Probabilities A	Payouts A	Probabilities B	Payouts B	EV(A)-EV(B)	OCI	Frac. chose A
1	0.25, 0.375, 0.375	-21, 0.5, 1.5	1	2	-6.5	0.06	4%
2	0.99, 0.01	18, 33	1	25	-6.9	0.10	9%
3	0.95, 0.05	-2, 28	0.75, 0.25	1, 22	-6.8	0.12	16%
4	0.75, 0.25	8, 52	1	26	-7	0.24	42%
5	0.8, 0.2	-30, 34	0.1, 0.9	7, -12	-7.1	0.27	43%
6	0.4, 0.3, 0.15, 0.15	-28, 31, 29, 25	1	7	-0.8	0.42	41%

5.1 Examples

Table 2 presents six example choice problems. Five of them are selected to have similar EV differences but varying *OCI*. All problems are such that Option A (the lottery with the weakly larger number of states) has a lower EV.

Problem 1 is very simple even though lottery A has three distinct payout states. This is because it features dominance (thus excess dissimilarity is 0), but also because the payout scale is relatively low. To see the role of excess dissimilarity more clearly, consider problems 2 and 3. In both problems, there is no dominance but the problems are intuitively simple. The reason is that excess dissimilarity is very low because the payout probabilities in lottery A are relatively extreme and the alternative is a safe payment. For instance, in problem 2, one can intuitively see that lottery A “is worth approximately \$18,” which makes B’s payout of \$25 look transparently superior.

In contrast, in problems 4 and 5, excess dissimilarity is high because the options have different advantages and disadvantages. For example, in problem 5, heuristic pairwise comparisons of the lottery upsides and downsides is difficult.

Finally, problem 6 illustrates a very high complexity problem in which the drivers of complexity are high excess dissimilarity, large support, a presence of losses, and also very similar EVs.

5.2 Choice Complexity, Behavioral Attenuation and Noise

A first potential signature of complexity – highlighted by a recent literature – is a form of attenuation, according to which decisions become less elastic to variation in

problem fundamentals. The top left panel of Figure 4 shows choice rates for lottery A as a function of the EV difference between A and B, separately for choice problems that are above or below median *OCI*. We label the lotteries such that lottery A is always the one with a weakly larger number of distinct payout states (lottery B is often a safe payment).

The figure shows a binned scatter plot because of the large number of underlying choice problems ($N = 10,898$). Each dot represents an equal number of choice problems (about 109 problems per dot), and show the average fraction of subjects choosing lottery A across all of the choice problems in a bin. We see that choice rates in problems that are predicted to be more complex are substantially more compressed towards 50% and, hence, more attenuated.¹⁸

A potential concern is that these differences between high- and low-complexity problems are confounded by the fact that the appropriate x-axis is not the EV difference but the true expected utility difference. To gauge this, the top right panel shows analogous results by plotting empirical choice rates against the estimated value difference in a cumulative prospect theory (CPT) model.¹⁹ The results are very similar.

The middle panels provide a complementary perspective, by showing choice rates for lottery A as a function of the continuous *OCI* index. In the left panel, the red dots correspond to cases where $EV(A) > EV(B)$ and the blue dots to cases where $EV(A) < EV(B)$. The middle right panel is constructed analogously but splits by CPT value difference. Thus, for an EV (or CPT) maximizer, the choice rates should be 0% and 100%. More interestingly, even standard random choice models would predict that choice rates are constant in *OCI*. Instead, we see that choice rates monotonically approach 50% as complexity increases. The magnitude of this effect is very large: choice rates for the option with the higher EV decrease by about 40 percentage

¹⁸One interpretation of the compression towards choice rates of 50-50 is that subjects have a prior belief over the expected utility associated with Options A and B, and that this prior is uninformative. We ran additional pre-registered experiments that study the role of prior beliefs and how they interact with problem complexity. In these experiments, we experimentally manipulate prior beliefs over which option is “better”. This treatment has a larger effect on choice for more complex problems. To conserve space, these experiments are summarized in Appendix F of our earlier NBER working paper (#31677).

¹⁹As discussed in detail in Section 6 and Appendix D, we estimate CPT as a representative agent model with the specifications that – according to a recent meta-analysis (Brown et al., 2024, Table 3) – are most common in the literature: a reference point of zero and CRRA utility.

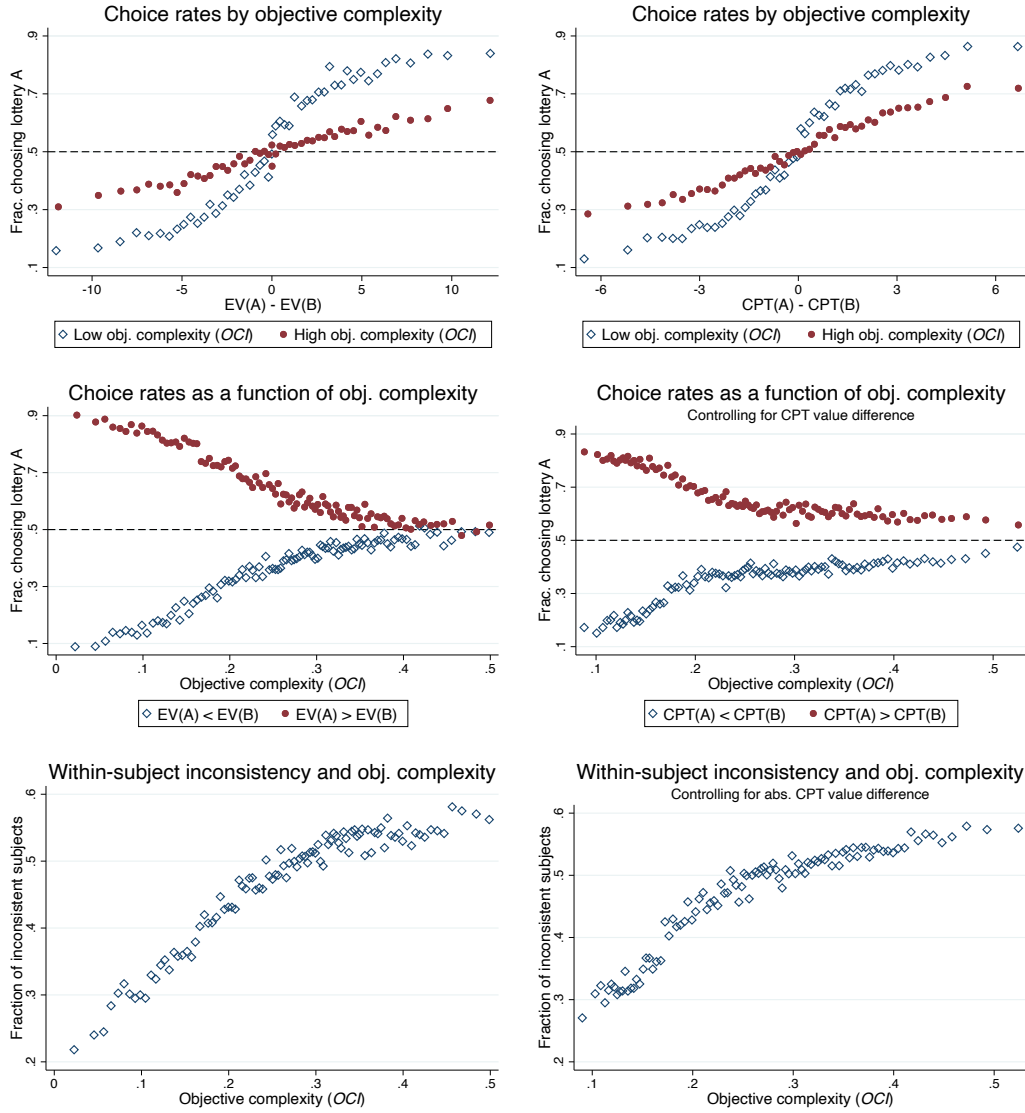


Figure 4: Objective complexity and behavioral attenuation in lottery choice. In the top and middle panels, the y-axis is the fraction of subjects choosing lottery A. The top panels implement a median split by *OCI*, separately within each percentile of the EV difference (or the estimated CPT value difference) between A and B. The middle panels split the choice problems according to whether A or B has a higher EV (or a higher estimated CPT value difference). In the bottom panels, the y-axis shows the fraction of subjects who are inconsistent at least once. The middle right panel controls for the estimated CPT value difference and its square, and the bottom right panel for the absolute estimated CPT value difference and its square. All panels show binned scatter plots. Top panels constructed from 10,898 choice problems, middle and bottom panels omit problems with absolute EV difference of less than \$0.20, hence constructed from 10,366 choice problems.

points going from very low to very high complexity. These results hold regardless of whether or not we control for the EV difference (or the estimated CPT value difference). This suggests that the vast majority of the explanatory power of *OCI* for behavioral attenuation is due to aggregation complexity (chiefly excess dissimilarity) rather than proximity to indifference. We further quantify this point below.

Noise or stable heuristics? Attenuation could arise through at least two different mechanisms. First, *OCI* could pick up noisy evaluations, producing attenuation akin to the effects in random choice models. Second, *OCI* could predict the use of stable heuristics. To study mechanisms, we link *OCI* to a standard measure of random choice: across-trial variability in repetitions of the same problem (within-subject choice inconsistencies). Recall that in Peterson et al. (2021), each subject that completed any given choice problem did so five times (consecutively). For each choice problem, we compute the fraction of subjects who are inconsistent at least once. In these analyses, we restrict attention to problems in which the absolute EV difference is at least \$0.20 to reduce concerns that “inconsistencies” simply reflect indifference (the results are identical in the full sample).

The bottom panels of Figure 4 show binned scatter plots of problem-level choice inconsistencies against *OCI*, either the raw correlation (left panel) or controlling for the absolute estimated CPT value difference (right panel). Moving from $OCI = 0$ to $OCI = 0.5$ is associated with an increase in the frequency of choice inconsistencies of 35 percentage points. The raw and partial correlations are around $r \approx 0.49 - 0.53$ ($p < 0.01$).²⁰

Benchmarking. A natural question is how quantitatively important complexity is for understanding choice behavior. Here, a natural point of comparison is the proximity to indifference (in the choice task), which is a main driver of choice errors in standard random choice models. We, hence, compare the predictive power of *OCI* with that of the subjective value difference in a CPT model, estimated as described in Appendix D. Because *OCI* includes the proximity of the EVs of the two options, we also benchmark *OCI* against the EV difference.

For our benchmarking analysis, we desire proxies for the frequency of choice errors. We work with three such proxies: (i) the fraction of subjects who are inconsistent at least once in repetitions of the same problem; (ii) average self-reported *CU* in a choice problem; and (iii) the fraction of decisions that does not correspond to choosing the lottery that – according to an estimated CPT model – delivers higher value. A caveat

²⁰ *OCI* and excess dissimilarity still strongly predict choice inconsistencies when dominance problems (the special case of minimal excess dissimilarity) are excluded from the sample, $r = 0.29$ for *OCI* and $r = 0.37$ for log excess dissimilarity ($p < 0.01$ for both).

Table 3: Benchmarking *OCI* and proximity to indifference

	<i>Dependent variable:</i>								
	Frac. inconsistent			Deviation from CPT prediction			Average <i>CU</i>		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>OCI</i>	0.70*** (0.01)			0.81*** (0.01)			50.0*** (3.97)		
Abs. EV diff.		-0.0078*** (0.00)			-0.017*** (0.00)			0.039 (0.18)	
Abs. CPT value diff.			-0.017*** (0.00)			-0.045*** (0.00)			0.17 (0.32)
Constant	0.27*** (0.00)	0.50*** (0.00)	0.50*** (0.00)	0.12*** (0.00)	0.41*** (0.00)	0.44*** (0.00)	2.94*** (0.93)	15.6*** (0.99)	15.4*** (0.84)
Observations	10398	10398	10398	10898	10898	10898	500	500	500
R^2	0.26	0.02	0.03	0.32	0.11	0.21	0.24	0.00	0.00

Notes. OLS estimates, robust standard errors in parentheses. An observation is a choice problem. The dependent variable in columns (1)–(3) is the fraction of subjects who are inconsistent at least once in the five repetitions of the choice problem. In columns (4)–(6) it is the fraction of decisions that does not equal the prediction of a full CPT model, see Appendix D. In columns (7)–(9) the dependent variable is average self-reported *CU* in the choice experiments (in percent). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

on the latter is that we estimate a representative agent CPT model that does not allow for across-subject heterogeneity.

Table 3 shows the results. The main metric of interest is the variance explained in each regression. The main takeaway is that *OCI* explains a considerably larger fraction of the variation than the EV difference or the estimated CPT value difference. For example, in column (6), the estimated value difference in a CPT model almost by construction explains a sizable share (21%) of the frequency of decisions that do not maximize CPT value. Yet the variance explained by *OCI* is substantially larger (36%, column (4)). Similarly, *OCI* strongly predicts *CU*, while the estimated distance to indifference does not predict it at all. These results again show that the vast majority of the predictive power of *OCI* reflects the difficulty of aggregating the constituent components of a lottery choice problem, rather than proximity to indifference.

5.3 Separate Lottery Features

Up to this point, we focused on the overall complexity of a choice set. We now disaggregate the previous results in two ways. First, we study the complexity of individual lotteries rather than of a choice set as a whole. This allows us to study not only behavioral attenuation but also potential complexity aversion. Second, we analyze the separate features that jointly make up the complexity index. After all,

given that excess dissimilarity drives the vast majority of the variation in *OCI*, it is not at all obvious that the results emphasized earlier – behavioral attenuation and choice noise – apply to every feature that is a part of the index.

To transparently study the complexity of individual lotteries, we restrict attention to decisions between a lottery (A) and a safe payment (B). This can be thought of as fixing the complexity of B at zero and varying the complexity of A.

We here focus on three lottery features given their disproportionate importance and prominence in the literature: (i) lottery variance because, as discussed in Section 4.2, it is the main driver of excess dissimilarity when the alternative is a safe payment; (ii) lottery support; and (iii) compound probabilities.²¹

Lottery variance: Behavioral attenuation and small-stakes risk aversion.

The left panel of Figure 5 shows a binned scatter plot of choice rates for the lottery as a function of the difference between the estimated CPT values of the lottery and the safe payment. To study the role of complexity, we now split by median lottery variance. In this plot, aversion to variance is indicated by a *uniform downward shift* of choice fractions. Thus, if people are risk averse and the degree of attenuation was independent of variance, we should expect choice rates for the high-variance lotteries to be lower than for the low-variance lotteries everywhere. Complexity-linked attenuation, on the other hand, is again indicated by a compression (or “flipping”) pattern, according to which observed risk taking can even increase in the lottery’s variance when the lottery is very unattractive (the left part of the figure).

In the data, we indeed see a pronounced compression pattern. This implies that people *look* risk averse when the lottery is attractive (to the right of zero), but risk loving when the lottery is unattractive (to the left of zero).

To highlight the confound this poses for estimating risk preferences, we restrict attention to the 2,678 problems for which all payouts are weakly positive (and one option is a safe payment), where we can estimate a standard CRRA expected utility model, $EU(x) = E[x^\alpha]$. When we estimate this model on the sub-sample in which the lottery has a higher EV than the safe payment, we estimate $\hat{\alpha} = 0.77$ (*s.e.* =

²¹Appendix Figure 12 shows choice rates split by the other two lottery-specific features in *OCI*, i.e., lottery scale and whether losses are present.

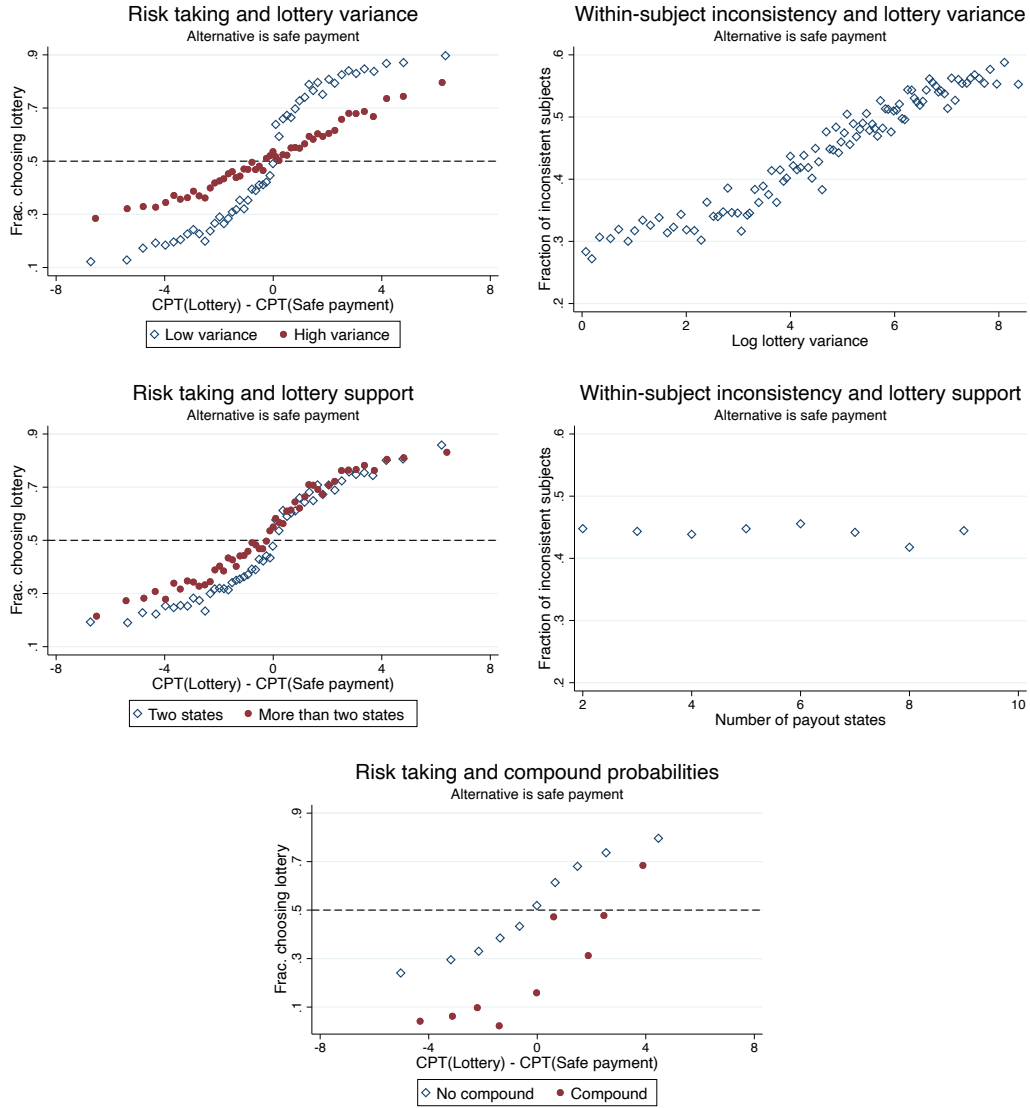


Figure 5: Binned scatter plots of choice rates (left panels) and choice inconsistency (right panels) for problems that involve one non-degenerate lottery and one safe payment. In the left panels, the y-axis is the fraction of subjects choosing lottery A. These panels implement a median split by lottery variance (separately within each percentile of the estimated CPT value difference) or by the number of payout states. In the right panels, the y-axis shows the fraction of subjects who are inconsistent at least once. Left panels constructed from 6,476 choice problems, right panels omit problems with absolute EV difference of less than \$0.20, hence constructed from 6,137 choice problems.

0.01) – a typical estimate suggesting small-stakes risk aversion. In contrast, when we estimate on the sub-sample in which the lottery has a lower EV than the safe payment, we estimate $\hat{\alpha} = 1.05$ ($s.e. = 0.01$) – if anything, suggesting apparent risk loving preferences.²² This exercise shows that complexity-linked attenuation can

²²We get similar results when we estimate a CARA model instead, $EU(x) = E[-e^{-\gamma x}]$. In the sub-sample of problems in which the lottery has a higher EV than the safe payment, we estimate $\hat{\gamma} = 0.0093$ ($s.e. = 0.0004$), and in the sub-sample in which the lottery has a lower EV than the safe payment, we estimate $\hat{\gamma} = -0.0008$ ($s.e. = 0.0003$).

predictably bias the estimation of preference parameters.

What is more, we can again provide evidence on the underlying mechanism. The top right panel of Figure 5 shows that lottery variance strongly predicts choice randomness (within-subject choice inconsistencies). Thus the top left and top right panels paint a consistent picture: as variance increases, choice becomes more random (and hence more compressed to 50-50).²³ This also implies that we see closely related patterns at two different levels of aggregation: the left panel (aggregate analysis) shows a compression or attenuation effect, while the right panel shows the link with choice inconsistencies at the individual level.

We do not argue that genuine small-stakes risk aversion does not exist. Rather, the point is that variance-linked attenuation is so strong that it can either amplify or entirely override any true aversion that likely exists.

Lottery support. The middle left panel of Figure 5 again shows choice rates for the lottery (when the alternative is a safe payment) as a function of the estimated CPT value difference, now split by whether the lottery has two or more payout states. Again, aversion to complexity would be indicated by a horizontal downward shift, while attenuation is indicated by a compression effect.

If anything, we see mild evidence of a compression effect and / or complexity seeking behavior. These patterns do not hinge on the specific sample split but look very similar when we instead split the sample at 3, 4 or 5 states. One reason for this null result is plausibly the mechanical effect of a heuristic of treating all payout states roughly equally (i.e., probability weighting). Intuitively, as is well-known in the literature (e.g. Wakker, 2022), if people treat all payout states the same to some degree, splitting up payout states will either produce seeming complexity aversion or complexity love, purely depending on whether the split states are attractive or unattractive (relative to the other payout states of the respective lottery).²⁴

²³An alternative interpretation of the compression pattern in the top left panel of Figure 5 is that subjects evaluate the lottery using CPT preferences and use the safe payment as a reference point, in which case prospect theory would predict the results. However, this does not explain the strong effect of lottery variance on random choice, which mechanically generates the compressed choice patterns. Moreover, this alternative explanation is conceptually unattractive because – as shown in a recent meta-analysis (Brown et al., 2024, Table 3) – prospect theory is almost always specified by assuming a reference point of zero or of the expected value across all options (rather than of the alternative).

²⁴Indeed, as we unpack in Appendix Table 9, we find that lottery support has a positive effect

The middle right panel shows the lack of a correlation between the number of payout states and choice inconsistencies ($r = -0.01$, $p = 0.32$). As documented by Arrieta and Nielsen (2023), one potential reason for this is that a larger number of payout states leads to a decision process that is more describable (procedural), and thus potentially less noisy.

Compound probabilities. The bottom panel again shows the familiar figure for choice rates, now split by whether the lottery involves compound probabilities. Here, we see a strong and statistically significant complexity aversion effect – for any given estimated CPT value difference, choice rates for the lottery are considerably smaller when it involves compound probabilities. We cannot consider the link between compound probabilities and choice inconsistencies because the PEA experiment did not include compound probabilities.

6 Structural Estimations

The reduced-form analysis in Section 5 provided evidence for both complexity-linked attenuation and complexity aversion. A natural question is thus how complexity can be incorporated into structural estimation to quantify magnitudes.

Incorporating complexity aversion into structural analyses is relatively straightforward, by adding a cost function to the decision maker’s objective (e.g., Puri, 2025). To incorporate complexity-driven attenuation, we allow the responsiveness parameter (or error variance) in a random choice model to depend on complexity. Suppose that choice probabilities are given by the logit model:

$$P(A) = F(EU(A) - EU(B); \eta) = \frac{1}{1 + e^{-\eta[EU(A) - EU(B)]}} \ , \quad (6)$$

where η is the conventional responsiveness (precision) parameter. In this model, unlike in the reduced-form *OCI* index developed earlier, aggregation complexity and resulting attenuation (captured by η) can be separated from proximity to indifference.

on choice rates for the lottery when the low-probability events have relatively high payouts, yet a negative effect when the low-probability events have relatively low payouts. The overall effect of weak ‘complexity seeking’ in Figure 5 may thus simply reflect that in our dataset low-probability events tend to have higher payouts, on average.

Appendix D develops a ‘structural’ complexity index that allows researchers to let the precision η in their estimated logit model to depend on complexity. This index of objective aggregation complexity (OAC) captures the predicted logit noisiness in the EV task, where the prediction is computed as a function of the same choice set features as OCI . Structural analyses can then be implemented by estimating a complexity-augmented logit model, in which the precision is heteroscedastic and specified as

$$\eta_{C,D} = \eta_0 + \eta_1/OAC_{C,D} + \epsilon_{C,D} , \quad (7)$$

where we supply $OAC_{C,D}$ and the researcher estimates η_0 and η_1 .

This exercise relies on the same logic and identifying assumptions as the complexity indices discussed above, except that we need to invoke the additional assumption that complexity impacts choice through the responsiveness parameter in a logit model.

We estimate eq. (6) and (7) at the population level using maximum likelihood for different combinations of (i) the specification of the DM’s objective function and (ii) the presence of complexity-dependent heteroscedasticity (i.e., whether η_1 is estimated or forced to be zero). Appendix D presents the estimating equation for each model as well as the resulting parameter estimates.

To start out, consider CPT. For each choice problem, the left panel of Figure 6 plots the actual choice rate for the lottery that has higher value (according to an estimated CPT model), as a function of OAC . In addition, we plot the model-predicted choice rates in a CPT model. As is clear from this figure, CPT has highly systematic prediction errors in our data: it strongly underpredicts how often people choose the estimated higher value option when complexity is low but overpredicts it when complexity is high.

The right panel of Figure 6 shows the prediction errors of a CPT model augmented by a complexity-dependent precision term. We see that predicted and actual choice rates track each other much more closely.

To systematically assess model fit, Figure 7 plots the variance explained across each of six models. To assess model completeness, the dashed horizontal line corresponds to the variance explained of a machine learning ensemble ($R^2 = 73\%$).

The first model assumes EV maximization and only estimates a constant precision

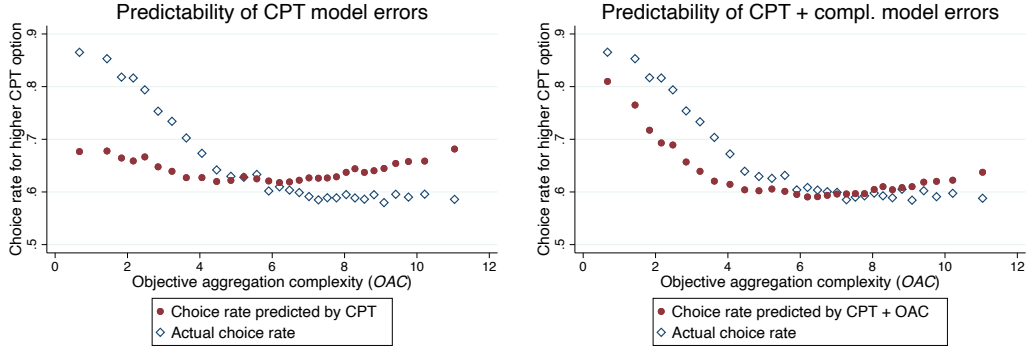


Figure 6: Model prediction errors in pooled choice data as a function of objective aggregation complexity (10,898 choice problems). Left panel: CPT model. Right panel: CPT model with complexity-dependent precision parameter. Both panels plot the actual and predicted choice rates for the choice option that the respective model predicts has higher CPT value.

parameter. The second model estimates CPT and includes separate utility curvature parameters for gains and losses (and hence includes reference-dependence relative to a reference point of zero), loss aversion and probability weighting. The third and fourth models are identical to the first two except that they also estimate the parameter that maps the complexity index into logit precision. The fifth and sixth models are identical to the first two except that they also estimate two complexity aversion parameters, one for compound lotteries and one for support.

The EV model has an R-squared of 44%, which increases to 57% in the full CPT model. Introducing one parameter that maps problem complexity into logit precision brings an EV model to $R^2 = 58\%$, slightly larger than the full CPT model. In other words, in our dataset, complexity-dependent attenuation alone is quantitatively more important than reference dependence, utility curvature, loss aversion and probability weighting combined. The variance explained further increases to 68% under the full complexity-dependent CPT specification.²⁵

Complexity aversion has smaller effects on model fit. Consistent with the discussion in Section 5.3, we usually estimate significant aversion against compound lotteries but not against lotteries with many states (see Appendix Table 10 for parameter estimates). The small increase in model fit (about 3% relative to CPT) is

²⁵These estimations assume CRRA utility and that the reference point is zero. Appendix Figure 15 shows that we get similar results when we assume CARA utility or a reference point that is given by the expected value of all lotteries in an experiment. Regardless of how we specify CPT, we always find that the variance explained increases substantially when the logit precision term is allowed to depend on complexity.

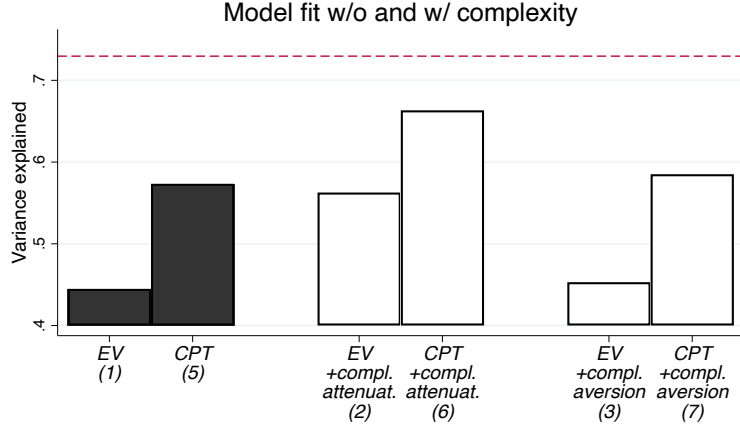


Figure 7: Variance explained of different choice models. Number of estimated model parameters in parentheses. R^2 is computed by first estimating each model, then using estimated model parameters to predict choice rates, and then regressing actual on predicted choice rates in the test set. The first model assumes EV maximization and a constant logit responsiveness term. The second model adds utility curvature parameters for gains and losses, loss aversion and probability weighting. The third and fourth models are analogous except that they also estimate the parameter that maps OAC into logit responsiveness, see eq. (7). The fifth and sixth models are identical to the first two except that they also allow for aversion to compound lotteries and lottery support. See Appendix D for all estimating equations and estimated parameter values. For all models, we show the performance in a test set of 2,726 choice problems after the models were estimated in a train sample of 8,172 choice problems. The dashed line corresponds to the performance of a machine learning ensemble.

partly driven by the construction of the dataset (only 0.4% of all choice problems involve compound lotteries). More generally, our results of course do not rule out that complexity aversion may dominate in other choice formats or that there are other features with respect to which complexity aversion may dominate.

We conclude from this analysis that allowing for complexity-linked attenuation is quantitatively important. This resonates with a literature in psychology that finds that allowing the noise term in a stochastic choice model to depend on lottery dissimilarity ($\delta_{A,B}$ rather than $\ln(d_{A,B})$) yields the best model fit relative to other models proposed in the psychology literature (Erev et al., 2010).

7 Discussion

This paper has made two contributions. First, we developed simple and interpretable indices of objective and subjective lottery choice complexity. A significant practical advantage of these indices is that they can be computed for any standard lottery choice dataset. A single feature – the excess dissimilarity that captures the tradeoff

complexity between the lotteries in a set – captures the bulk of variation in complexity. We have emphasized that our indices are best understood as a *predictive* tool that allows researchers to quantify complexity and predict behavior, rather than as a *causal* identification device. Our second contribution is to comprehensively study behavioral signatures of complexity, which also allows us to illustrate the large predictive power of the complexity indices.

Larger menus. A natural question is how our indices can be applied to larger choice sets. In our *EV Tasks* experiment, we also included menus with between three and five options. Appendix F discusses the results. Menu size is strongly linked to both error rates and cognitive uncertainty. This suggests that incorporating menu size into our indices would be productive. The main challenge we see is that extending to larger menus would necessitate generalizing measures such as excess dissimilarity to larger choice sets (see Natenzon (2019) and Shubatt and Yang (2024) for steps in this direction).

Other decision domains? A natural question is whether the approach we deploy in this paper could also be used to quantify the complexity of other economic decisions. Our approach requires an objective task that is somewhat “close” to the choice problem of interest. It thus likely cannot be used to, say, quantify the complexity of choosing between two cars.

However, we believe that there are other settings in which our approach could add valuable insights. For instance, Guan et al. (2025) use techniques similar to ours to quantify the complexity of information choice problems. Similarly, one could develop a composite index of intertemporal choice complexity using the same techniques as in this paper. Here, the objective task would consist of a present value task – as in Enke et al. (2025), the researcher could endow subjects with some discount function (say, $D(t) = 0.96^t$) and then ask them to identify the payment with the higher present value. Using this technique, a composite complexity index could be developed from mistake rates, and subsequently deployed to predict intertemporal choices over money.

References

- Agranov, Marina and Pietro Ortoleva**, “Stochastic choice and preferences for randomization,” *Journal of Political Economy*, 2017, 125 (1), 40–68.
- Andersson, Ola, Håkan J Holm, Jean-Robert Tyran, and Erik Wengström**, “Risk aversion relates to cognitive ability: Preferences or noise?,” *Journal of the European Economic Association*, 2016, 14 (5), 1129–1154.
- Arrieta, Gonzalo and Kirby Nielsen**, “Procedural Decision-Making In The Face Of Complexity,” *Working Paper*, 2023.
- Arts, Sara, Qiyang Ong, and Jianying Qiu**, “Measuring decision confidence,” *Experimental Economics*, 2024, pp. 1–22.
- Bastianello, Francesca and Alex Imas**, “Biases in belief updating within and across domains,” *Available at SSRN 5413920*, 2025.
- Benjamin, Daniel J, Sebastian A Brown, and Jesse M Shapiro**, “Who is ‘behavioral’? Cognitive ability and anomalous preferences,” *Journal of the European Economic Association*, 2013, 11 (6), 1231–1255.
- Bernheim, B Douglas and Charles Sprenger**, “On the empirical validity of cumulative prospect theory: Experimental evidence of rank-independent probability weighting,” *Econometrica*, 2020, 88 (4), 1363–1409.
- Birnbaum, Michael H**, “Three new tests of independence that differentiate models of risky decision making,” *Management Science*, 2005, 51 (9), 1346–1358.
- Brown, Alexander L, Taisuke Imai, Ferdinand M Vieider, and Colin F Camerer**, “Meta-analysis of empirical estimates of loss aversion,” *Journal of Economic Literature*, 2024, 62 (2), 485–516.
- Bruhin, Adrian, Helga Fehr-Duda, and Thomas Epper**, “Risk and rationality: Uncovering heterogeneity in probability distortion,” *Econometrica*, 2010, 78 (4), 1375–1412.

- Buschena, David and David Zilberman**, “Generalized expected utility, heteroscedastic error, and path dependence in risky choice,” *Journal of Risk and Uncertainty*, 2000, 20, 67–88.
- Camerer, Colin and Teck-Hua Ho**, “Violations of the betweenness axiom and nonlinearity in probability,” *Journal of Risk and Uncertainty*, 1994, 8, 167–196.
- Carvalho, Leandro and Dan Silverman**, “Complexity and sophistication,” Technical Report, National Bureau of Economic Research 2019.
- Enke, Benjamin and Thomas Graeber**, “Cognitive uncertainty,” *The Quarterly Journal of Economics*, 2023, 138 (4), 2021–2067.
- , —, and **Ryan Oprea**, “Confidence, self-selection and Bias in the Aggregate,” *American Economic Review*, 2023.
- , —, and —, “Complexity and time,” *Journal of the European Economic Association*, 2025, p. jvaf009.
- , —, —, and **Jeffrey Yang**, “Behavioral Attenuation,” *Working Paper*, 2024.
- Erev, Ido, Alvin E Roth, Robert Slonim, and Greg Barron**, “Combining a theoretical prediction with experimental evidence,” *Available at SSRN 1111712*, 2002.
- and **Greg Barron**, “On adaptation, maximization, and reinforcement learning among cognitive strategies,” *Psychological review*, 2005, 112 (4), 912.
- , **Eyal Ert, Alvin E Roth, Ernan Haruvy, Stefan M Herzog, Robin Hau, Ralph Hertwig, Terrence Stewart, Robert West, and Christian Lebiere**, “A choice prediction competition: Choices from experience and from description,” *Journal of Behavioral Decision Making*, 2010, 23 (1), 15–47.
- , —, **Ori Plonsky, Doron Cohen, and Oded Cohen**, “From anomalies to forecasts: Toward a descriptive model of decisions under risk, under ambiguity, and from experience,” *Psychological review*, 2017, 124 (4), 369.

- Fishburn, Peter C**, “A probabilistic expected utility theory of risky binary choices,” *International Economic Review*, 1978, pp. 633–646.
- Frydman, Cary and Lawrence Jin**, “Efficient Coding and Risky Choice,” *Quarterly Journal of Economics*, 2021.
- Fudenberg, Drew and Indira Puri**, “Evaluating and extending theories of choice under risk,” *Working paper, MIT Economics*, 2021.
- and —, “Simplicity and probability weighting in choice under risk,” in “AEA Papers and Proceedings,” Vol. 112 2022, pp. 421–425.
- , **Jon Kleinberg, Annie Liang, and Sendhil Mullainathan**, “Measuring the completeness of economic models,” *Journal of Political Economy*, 2022, 130 (4), 956–990.
- Gabaix, Xavier and Thomas Graeber**, “The Complexity of Economic Decisions: Theory and Experimental Evidence,” *Working Paper*, 2023.
- Georgalos, Konstantinos and Nathan Nabil**, “Testing models of complexity aversion,” *Journal of Behavioral and Experimental Economics*, 2025, 116, 102354.
- Gillen, Ben, Erik Snowberg, and Leeat Yariv**, “Experimenting with Measurement Error: Techniques with Applications to the Caltech Cohort Study,” *Journal of Political Economy*, 2019.
- Guan, Menglong, Ryan Oprea, and Sevgi Yuksel**, “Beyond Instrumental Value: How Complexity Shapes Information Demand,” *Working Paper*, 2025.
- Halevy, Yoram**, “Ellsberg revisited: An experimental study,” *Econometrica*, 2007, 75 (2), 503–536.
- He, Junnan and Paulo Natenzon**, “Moderate utility,” *Available at SSRN 3243657*, 2022.
- Hey, John D**, “Experimental investigations of errors in decision making under risk,” *European Economic Review*, 1995, 39 (3-4), 633–640.

- **and Chris Orme**, “Investigating generalizations of expected utility theory using experimental data,” *Econometrica: Journal of the Econometric Society*, 1994, pp. 1291–1326.
- Hu, En Hua**, “A Procedural Model of Complexity Under Risk,” *Working Paper*, 2023.
- Huck, Steffen and Georg Weizsäcker**, “Risk, complexity, and deviations from expected-value maximization: Results of a lottery choice experiment,” *Journal of Economic Psychology*, 1999, *20* (6), 699–715.
- Iyengar, Sheena S and Emir Kamenica**, “Choice proliferation, simplicity seeking, and asset allocation,” *Journal of Public Economics*, 2010, *94* (7-8), 530–539.
- Loomes, Graham, Chris Starmer, and Robert Sugden**, “Observing Violations of Transitivity by Experimental Methods,” *Econometrica*, 1991, *59* (2), 425–439.
- Martínez-Marquina, Alejandro, Muriel Niederle, and Emanuel Vespa**, “Failures in contingent reasoning: The role of uncertainty,” *American Economic Review*, 2019, *109* (10), 3437–74.
- Mononen, Lasse Miiikka**, “Essays on Rationality: Modeling and Measurement.” PhD dissertation, Princeton University 2021.
- Natenzon, Paulo**, “Random choice and learning,” *Journal of Political Economy*, 2019, *127* (1), 419–457.
- Noussair, Charles N., Stefan T. Trautmann, and Gijs Van de Kuilen**, “Higher order risk attitudes, demographics, and financial decisions,” *Review of Economic Studies*, 2014, *81* (1), 325–355.
- Oprea, Ryan**, “Decisions under risk are decisions under complexity,” *American Economic Review*, 2024, *114* (12), 3789–3811.
- Peterson, Joshua C, David D Bourgin, Mayank Agrawal, Daniel Reichman, and Thomas L Griffiths**, “Using large-scale experiments and machine learning to discover theories of human decision-making,” *Science*, 2021, *372* (6547), 1209–1214.

- Puri, Indira**, “Simplicity and risk,” *The Journal of Finance*, 2025, *80* (2), 1029–1080.
- Shubatt, Cassidy and Jeffrey Yang**, “Similarity and Comparison Complexity,” *Working Paper*, 2024.
- Sonsino, Doron, Uri Benzion, and Galit Mador**, “The complexity effects on choice with uncertainty—Experimental evidence,” *The Economic Journal*, 2002, *112* (482), 936–965.
- Tversky, Amos and Daniel Kahneman**, “Advances in prospect theory: Cumulative representation of uncertainty,” *Journal of Risk and uncertainty*, 1992, *5* (4), 297–323.
- **and J Edward Russo**, “Substitutability and similarity in binary choices,” *Journal of Mathematical psychology*, 1969, *6* (1), 1–12.
- Vieider, Ferdinand M**, “Noisy coding of time and reward discounting,” Technical Report, Ghent University, Faculty of Economics and Business Administration 2021.
- , “Decisions under Uncertainty as Bayesian Inference on Choice Options,” *Working Paper*, 2022.
- Wakker, Peter P**, “A Criticism of Bernheim & Sprenger’s (2020) Tests of Rank Dependence,” *Available at SSRN 4188758*, 2022.
- Weaver, Warren**, “Science and complexity,” *American scientist*, 1948, *36* (4), 536–544.
- Webb, Ryan, Paul W Glimcher, and Kenway Louie**, “The normalization of consumer valuations: Context-dependent preferences from neurobiological constraints,” *Management Science*, 2021, *67* (1), 93–125.
- Yang, Jeffrey**, “On the Decision Relevance of Subjective Beliefs,” *Working Paper*, 2023.
- Zilker, Veronika, Ralph Hertwig, and Thorsten Pachur**, “Age differences in risk attitude are shaped by option complexity,” *Journal of Experimental Psychology: General*, 2020, *149* (9), 1644.

ONLINE APPENDIX

A Previous Literature

Table 4: Experimental literature on lottery complexity

Choice set feature	Result	Papers
Number of options	Prefer small support	Iyengar and Kamenica (2010), Carvalho and Silverman (2019)
Number of states	Aversion	Huck and Weizsäcker (1999), Sonsino et al. (2002), Bernheim and Sprenger (2020), Puri (2025), Fudenberg and Puri (2022)
Number of states	Seeking	Birnbaum (2005), Erev et al. (2017), see Wakker (2022) for additional references
Number of states	Higher noise	Hey (1995), Huck and Weizsäcker (1999), Sonsino et al. (2002), Zilker et al. (2020), Arts et al. (2024)
Number of states	More describable	Arrieta and Nielsen (2023), Georgalos and Nabil (2025)
Absolute dist. b/w CDFs	Higher noise	Buschena and Zilberman (2000), Erev et al. (2002), Erev et al. (2010)
Compound prob.	Aversion	Halevy (2007), Gillen et al. (2019)
Compound prob.	Higher noise	Enke and Graeber (2023)
Opaque payouts / prob.	Higher noise	Enke and Graeber (2023), Zilker et al. (2020)
Payout range	Higher noise	Bruhin et al. (2010)
Payout magnitude	Higher noise	Webb et al. (2021)
Dominance	Lower noise	Agranov and Ortoleva (2017)
Payout variance (decisions from experience)	Higher noise	Erev and Barron (2005)

B Derivation and Completeness of Complexity Indices

B.1 Screenshot of Interface

See Figure 8.

B.2 Potential Complexity Features

Consider a choice between two lotteries indexed by j and denoted by letters A , B etc. Each lottery is characterized by payout probabilities $(p_1^j, \dots, p_{k_j}^j)$ and payoffs

Which lottery has the highest average payout if the computer runs it many, many times?
Please select one.

Lottery A

Prob. 5%: Get \$26
 Prob. 10%: Get \$24
 Prob. 15%: Get \$23
 Prob. 10%: Get \$17
 Prob. 5%: Get \$11
 Prob. 15%: Get \$5
 Prob. 40%: Lose \$15

Lottery B

Prob. 100%: Get \$11

How certain are you that each lottery has the highest average payout?
Please allocate 100 certainty points.

points

points

100 points left to allocate.

Remember, the "Next" button will only appear if you selected a lottery and entered a number into each textbox!

Figure 8: Example decision screen in *EV Tasks*

$(x_1^j, \dots, x_{k_j}^j)$, where k_j denotes the number of distinct payout states of lottery j . In the construction of our complexity indices, we include the features listed in Table 5. Whenever a feature is defined for a single lottery rather than a choice set, we include the average feature in the set. For continuous features (and “number of states”) f , we include the linear term (f), square (f^2), and the natural log ($\ln(f + 0.1)$).

B.3 Derivation of Indices

Table 6 shows the results of the exploratory LASSO regressions.

Table 7 shows the coefficients in each of the handcrafted indices. We winsorize the indices from below at zero to and from above at 0.5 to facilitate their interpretation of predicted error rates / predicted cognitive uncertainty. OAC and SAC capture predicted logit noisiness and are thus only winsorized from below at zero.

B.4 Completeness

To assess completeness, we follow Fudenberg et al. (2022) and benchmark the out-of-sample performance of our complexity indices against that of a highly flexible non-parametric model. In particular, we construct an ensemble predictor from the predictions of (1) our objective complexity LASSO, (2) *OCI*, and (3) a fully-flexible

Table 5: Potential complexity features

Feature	Defined on	Formal definition
Number of states	Option	k_j
Payout range	Option	$\max\{x_1^j, \dots, x_{k_j}^j\} - \min\{x_1^j, \dots, x_{k_j}^j\}$
Variance	Option	$\sum_{s=1}^{k_j} p_s^j (x_s^j)^2 - \left(\sum_{s=1}^{k_j} p_s^j x_s^j\right)^2$
Payout variance	Option	$1/k_j \sum_{s=1}^{k_j} (x_s^j - \bar{x}^j)^2$
Probability variance	Option	$1/k_j \sum_{s=1}^{k_j} (p_s^j - \bar{p}^j)^2$
Magnitude	Option	$1/k_j \sum_{s=1}^{k_j} x_s^j $
Pure Gains	Option	$\mathbb{1}\{x_s^j \geq 0 \forall j\}$
Mixed	Option	$\mathbb{1}\{\exists x_s^j > 0 \wedge \exists x_s^j < 0\}$
Pure Loss	Option	$\mathbb{1}\{x_s^j < 0 \forall j\}$
Distance to certainty	Option	$1/k_j \sum_{s=1}^{k_j} \min\{p_s^j; 1 - p_s^j\}$
Payout-weighted dist. to certainty	Option	$1/k_j \sum_{s=1}^{k_j} x_s^j \min\{p_s^j; 1 - p_s^j\}$
Entropy	Option	$\sum_{s=1}^{k_j} p_s^j (-\ln(p_s^j))$
Normalized payout dispersion	Option	$1/k_j \sum_{s=1}^{k_j} \frac{ x_s^j - \bar{x}^j }{ \bar{x}^j }$
Normalized Variance	Option	$(1/\text{Magn.}^2) \cdot \text{Var.}$, with Magn., Var. as defined above
Irregular probabilities	Option	$\mathbb{1}\{p_s^j \notin \{0.01, 0.05, 0.1, \dots, 0.9, 0.95, 0.99\} \text{ for } s = 1, \dots, k_j\}$
CDF self-distance	Option	$\sum_{s=1}^{k_j} x_s^j - EV(j) p_s^j$
Compound	Option	
Compound Range	Option	Range of distribution of unknown p
Weak dominance	Choice set	$\mathbb{1}\{F_A(x) \leq F_B(x) \forall x \text{ or } F_B(x) \leq F_A(x) \forall x\}$
Excess dissimilarity	Choice set	$\int_{\mathbb{R}} F_A(x) - F_B(x) dx - EV(A) - EV(B) $
Average absolute payoff difference	Choice set	$1/k \sum_{s=1}^k x_s^A - x_s^B $
Probability difference	Choice set	$\sum_{x \in X} f_A(x) - f_B(x) $, where $X = \{x_1^A, \dots, x_{k_A}^A\} \cup \{x_1^B, \dots, x_{k_B}^B\}$

convolutional neural network (CNN). The CNN is trained to predict error rates in a non-parametric fashion based on the raw lottery features (payouts and probabilities) and the handcoded features that enter our LASSO regressions. We then form an ensemble predictor by using OLS regression to combine these three predictors in a validation set of 105 problems. We evaluate all models in a test set of 531 problems. Formally, completeness is defined as $C(\hat{y}) = \frac{R_y^2 - R_{\text{Prox}}^2}{R_{\text{Ens}}^2 - R_{\text{Prox}}^2}$ where R_y^2 is the variance explained by $\hat{y}$, R_{Ens}^2 is the variance explained by the ensemble predictor $\hat{y}_{\text{Ens}}$, and R_{Prox}^2 is the variance explained by a baseline model which depends only on proximity to indifference. Variance explained is always calculated in the test set. Figure 9 shows the variance in mistake rates explained by *OCI*. To comprehensively study completeness, we show the results for four different types of indices. First, an index that only captures proximity to indifference: the absolute difference of the expected

Table 6: LASSO coefficients for *OCI* and *SCI*

Feature	Coefficients		Feature	Coefficients	
	OCI	SCI		OCI	SCI
Intercept	1.5E-01	4.3E-02	Sq Scale	-2.6E-07	2.4E-06
Log Abs. EV Difference	-9.4E-02	-1.3E-02	Sq Range		1.3E-07
Scale		-1.0E-03	Sq Variance	3.9E-11	-5.0E-10
Range		-3.3E-04	Sq Payout Variance	9.3E-11	2.7E-10
Variance		1.6E-05	Sq Num States		9.1E-04
Pay-wtd DC	-1.7E-03		Sq DC		-1.1E-01
Probability Variance		-4.4E-05	Sq Pay-wtd DC		1.3E-06
Payout Dispersion	6.6E-03		Sq Entropy		-2.4E-02
CDF Self-Dist	-1.2E-03	-2.4E-03	Sq Prob. Variance		1.9E+00
Log Scale	3.8E-02	2.2E-02	Sq Payout Dispersion		6.1E-03
Log Range		1.7E-02	Sq Norm. Variance	6.3E-03	1.8E-02
Log Variance		-2.0E-02	Sq CDF Self Dist		1.3E-05
Log Payout Variance	1.7E-02		Gains	-9.3E-03	-1.9E-02
Log Num States	2.8E-02	-2.4E-02	Irregular Probabilities	4.0E-02	-2.3E-03
Log DC	-1.9E-01		Excess Dissimilarity		-1.2E-03
Log Pay-wtd DC	-7.5E-05	1.6E-02	Log Excess Dissimilarity	5.3E-02	3.5E-02
Log Entropy	-2.4E-02	1.4E-01	Dominance	-5.2E-02	-2.4E-02
Log Prob. Variance		-3.7E-01	Compound	5.5E-02	3.6E-02
Log Payout Dispersion		-8.8E-03	Compound Range	1.2E-01	1.0E-01
Log Norm. Variance		-1.5E-02	Safe Option	7.2E-03	2.0E-02
Log CDF Self-Dist		3.6E-02			

Notes. Coefficients of LASSO regression of problem-level errors rates or cognitive uncertainty on choice set features in the *EV Tasks* experiment. DC = distance to certainty. Features that apply to a single lottery (such as number of states) are averaged across the lotteries in the set. Only features with at least one non-zero coefficient are included.

values and its square. Second, indices that only consist of the separate aggregation complexity features, such as log excess dissimilarity. Third, our handcrafted *OCI* index. Fourth, the analogous exploratory LASSO indices.

There are three main takeaways. First, proximity to indifference performs substantially worse than our index. By construction, its completeness is 0%. Second, log excess dissimilarity explains three times as much variation as proximity to indifference (66.6% complete). Third, our handcrafted index is almost as complete as the exploratory LASSO index (95% and 98% complete respectively). We conclude that our index captures a large fraction of the predictable component of complexity.

C Details for and Analyses of Choice Experiments

C.1 Tables

Table 7: Coefficients of features in complexity indices

Index:	<i>Dependent variable:</i>			
	Error rate		Implied obj. logit imprecision s^{EV}	
	<i>OCI</i>	<i>SCI</i>	<i>OAC</i>	<i>SAC</i>
	(1)	(2)	(3)	(4)
Log excess dissimilarity	0.058*** (0.00)	0.022*** (0.00)	1.76*** (0.13)	1.86*** (0.14)
Average log payout magnitude	0.035*** (0.00)	0.0029 (0.00)	1.11*** (0.18)	1.41*** (0.20)
Average log number of states	0.044*** (0.02)	0.052*** (0.01)	2.06*** (0.59)	3.09*** (0.63)
Frac. lotteries involving loss	0.028*** (0.01)	0.027*** (0.00)	1.16*** (0.32)	2.23*** (0.35)
1 if involves compound prob.	0.072*** (0.02)	0.060*** (0.01)	2.45*** (0.63)	3.39*** (0.67)
Log absolute EV difference	-0.089*** (0.01)	-0.015*** (0.00)		
Constant	0.12*** (0.03)	0.067*** (0.01)	-3.56*** (0.77)	-4.65*** (0.83)
Observations	1587	1587	1587	1587
R^2	0.31	0.36	0.20	0.24

Notes. OLS estimates, robust standard errors in parentheses. An observation is a decision problem from the train set in the *EV Tasks* experiment. In columns (3) and (4), the dependent variable is the implied logit precision $s_{A,B}^{EV}$ as defined in eq. (10) (either defined through mistakes or through cognitive uncertainty). To calculate $s_{A,B}^{EV}$, we first winsorize the error rates and CU so that they never exceed 0.49. Then, we winsorize the calculated $s_{A,B}^{EV}$ at the 85th percentile. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

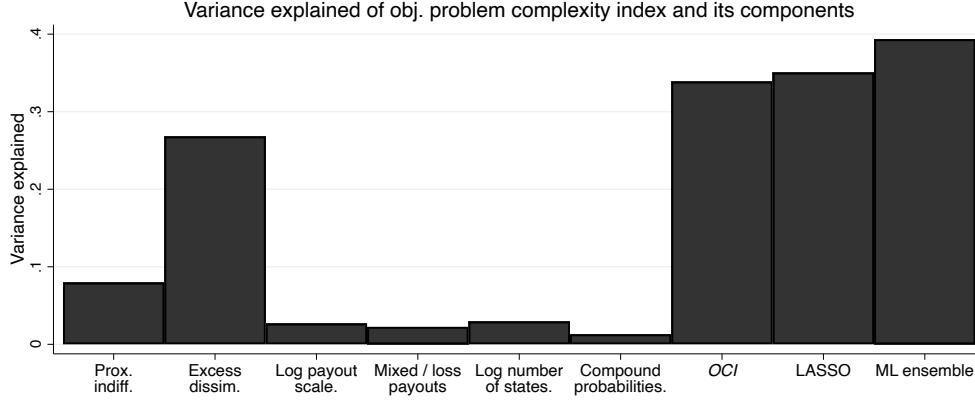


Figure 9: Variance explained of complexity indices and separate features. Each bar reports the variance explained of a regression of mistake rates in the EV task on the respective feature (in the test set). Leftmost model includes linear and squared terms of absolute expected values difference. Second model includes log excess dissimilarity, and so on. Seventh model is *OCI*, eight model the exploratory LASSO index and the ninth model the predictions of a machine learning ensemble.

Table 8: Summary statistics for problems across experiments

Experiment		# options	Safe payment	# states	Var	Scale	Mixed	Dominance
<i>EV Tasks</i>	Mean, median	2.1, 2	58%	3.3, 2	633, 164	34, 26	46%	5%
	Range	2, 5		2, 7	0, 39440	1, 213		
<i>Choice Tasks</i>	Mean	2	80%	3.3	775	25.5	49%	7%
	IQR	2, 2		2, 5	20, 735	14, 43		
<i>Choice Tasks from PEA</i>	Mean	2	59%	3.7	460	30	53%	17%
	IQR	2, 2		2, 5	20, 553	15, 40		

Notes. *PEA* = Peterson et al. (2021). For the *EV Task*, statistics are limited to problems with menu size two except for # options. Scale = absolute average payout. We display information for the lottery with the largest number of distinct payout states.

Table 9: Choice rates and lottery support

Sample: Payout difference:	<i>Dependent variable:</i>					
	Fraction of subjects choosing lottery (in %)					
	Full	≥ 0	< 0	< -10	< -20	< -30
	(1)	(2)	(3)	(4)	(5)	(6)
Support of lottery	0.97*** (0.08)	1.26*** (0.10)	0.34* (0.18)	-0.26 (0.30)	-1.79*** (0.44)	-2.97*** (0.55)
CPT value difference	6.67*** (0.07)	5.83*** (0.08)	7.97*** (0.12)	7.50*** (0.15)	6.42*** (0.20)	4.96*** (0.27)
Constant	48.8*** (0.36)	47.2*** (0.48)	50.8*** (0.60)	51.8*** (0.87)	55.7*** (1.20)	56.6*** (1.54)
Observations	6476	3725	2751	1682	866	410
R^2	0.59	0.57	0.64	0.60	0.56	0.49

Notes. OLS estimates, robust standard errors in parentheses. An observation is a choice problem. The sample consists of problems in which A is a non-degenerate lottery and B is a safe payment. The sample is split by the difference in average payouts between states that have low probability (defined as smaller than $1/N$) and states that have high probability (defined as larger than $1/N$), where N is the number of states of A. Positive values mean that low-probability states have relatively high payouts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Figures

Decision 1/50

Which lottery do you choose?
Please select one.

Lottery A

Probability 80%: **Get \$42**

Probability 20%: **Get \$4**

Lottery B

Probability 100%: **Get \$28**

How certain are you that you actually prefer the lottery you chose above?

Fully certain I prefer the lottery I didn't choose | Fully certain I prefer the lottery I chose

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

I am **PLEASE CLICK SLIDER** certain that I actually prefer the lottery I chose above.

Remember, the "Next" button will only appear if you selected a lottery and indicated your certainty on the slider!

Figure 10: Example decision screen in *Choice Tasks*

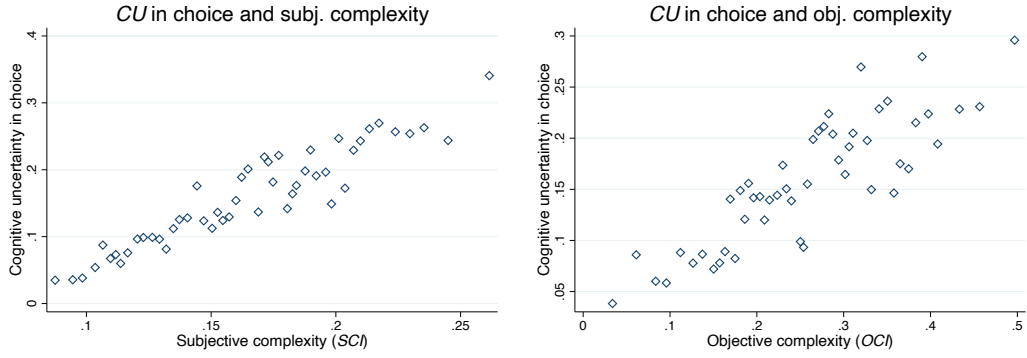


Figure 11: Average cognitive uncertainty in choice and complexity indices in experiment *Choice Tasks*. The figures show binned scatter plots of average cognitive uncertainty in each of 500 lottery choice problems, as a function of the predicted complexity indices. In the figure, an underlying observation is a choice problem. Figure is constructed from subjects who did not complete any EV task. Data from 250 subjects and 500 choice sets.

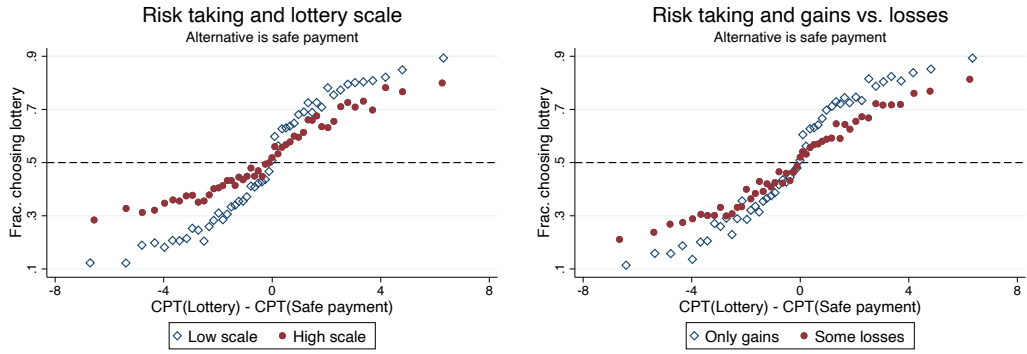


Figure 12: Binned scatter plots of choice rates for problems that involve one non-degenerate lottery and one safe payment. The y-axis is the fraction of subjects choosing lottery A. These panels implement a median split by lottery scale (separately within each percentile of the estimated CPT value difference) or by whether the lottery involved a loss. Constructed from 6,476 choice problems.

C.3 Replication of Figures 4 and 5 Excl. Losses in PEA data

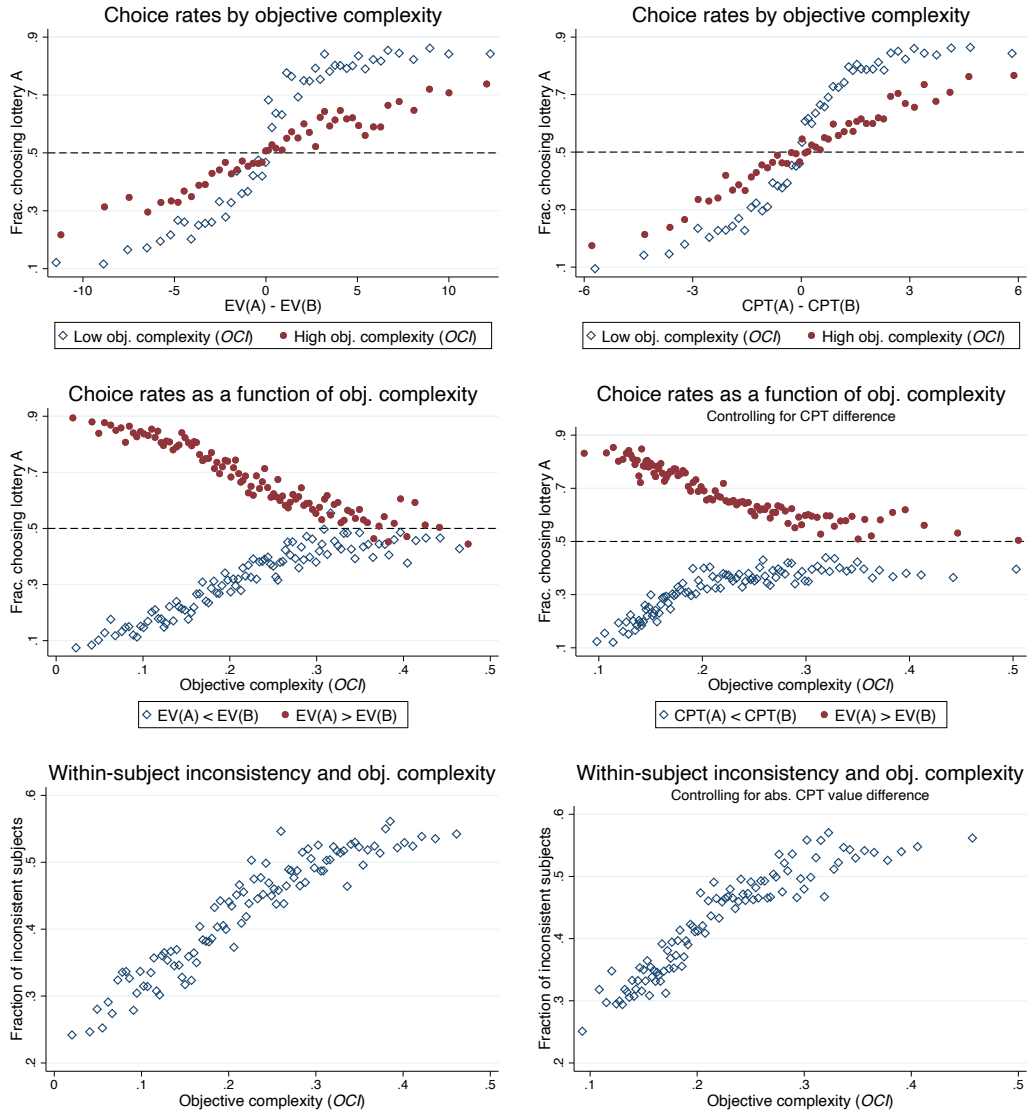


Figure 13: Replication of Figure 4, excluding choice problems in PEA dataset that involve losses.

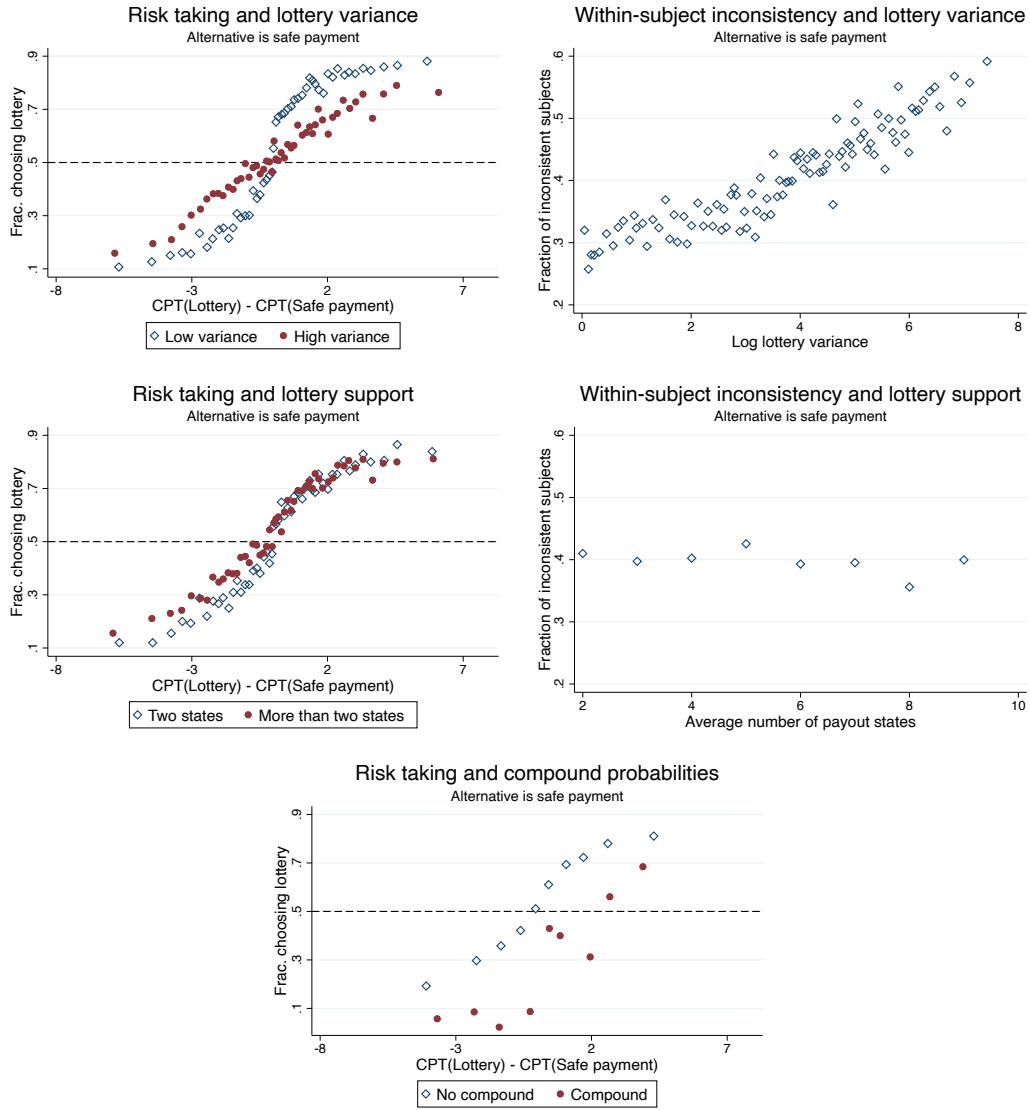


Figure 14: Replication of Figure 5, excluding choice problems in PEA dataset that involve losses.

D Details on Structural Estimations

D.1 Derivation of Structural Complexity Index

Suppose that choice probabilities are given by the logit model:

$$P(A) = F(EU(A) - EU(B); \eta) = \frac{1}{1 + e^{-\eta[EU(A) - EU(B)]}} , \quad (8)$$

where η is the conventional responsiveness (precision) parameter. We define the aggregation complexity of a choice problem as the inverse of the decision-maker's precision for that problem. Inverting the logit CDF,

$$s_{A,B} := \frac{1}{\eta_{A,B}} = \frac{EU(B) - EU(A)}{\ln\left(\frac{1}{P(A)} - 1\right)} . \quad (9)$$

Again, we cannot observe this object in choice data since we do not know the utility function. However, in the EV task, we can compute the “implied logit precision”²⁶ because we observe both the EV difference and the empirical rate of choosing A.

$$s_{A,B}^{EV} := \frac{1}{\eta_{A,B}^{EV}} = \frac{EV(B) - EV(A)}{\ln\left(\frac{1}{P(A)} - 1\right)} . \quad (10)$$

Similarly to before, we develop an index of predicted aggregation complexity by imposing the assumption that the problem-specific precision in the choice task is a linear function of the precision in the EV task,

$$\eta_{A,B} = \eta_0 + \eta_1 \frac{1}{s_{A,B}^{EV}} \quad \text{with} \quad \eta_1 > 0. \quad (11)$$

Definition 2. *The objective aggregation complexity ($OAC_{C,D}$) of choice set $\{C,D\}$ under a logit model is given by the prediction of the regression*

$$s_{A,B}^{EV} = \sum_{i=0}^N \alpha_i f_i^{A,B} + \epsilon_{A,B} . \quad (12)$$

²⁶Empirically, we winsorize selection rates for the higher EV lottery from below at 0.51 (since otherwise (9) is undefined). Next, we winsorize the across-problem distribution of $\hat{s}_{A,B}^{EV}$ at the 85th percentile because $s_{A,B}^{EV}$ can explode when selection rates for the wrong lottery is close to 50% or when the expected values difference is very small.

where $s_{A,B}^{EV}$ is calculated as in (10). Thus, the index is $OAC_{C,D} = \hat{s}_{C,D}^{EV} = \sum_{i=0}^N \hat{\alpha}_i f_i^{C,D}$.

We define an analogous index of subjective aggregation complexity (*SAC*), in which objective mistakes rates in the EV task are replaced with subjective ones (*CU*).

D.2 Model Estimation

Cumulative prospect theory. We allow up to four additional parameters: loss aversion with respect to a reference point of zero, separate utility curvature for gains and losses, and probability weighting:

$$EU_{CPT}(x) = \sum_i \pi_i u(x_i) , \quad (13)$$

where

$$u(x) = \begin{cases} x^\alpha & \text{if } x \geq 0 \\ -\lambda(-x)^\beta & \text{if } x < 0 \end{cases} \quad (14)$$

and π_i is a Tversky and Kahneman (1992) cumulative prospect theory decision weight with capacity function

$$w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}} . \quad (15)$$

Complexity aversion / seeking. In some specifications, we allow the value function to incorporate as-if complexity preferences. Based on earlier results, we allow for aversion (or seeking) responses to the number of states in the lottery and whether the lottery is compound. For these models, we let the value function take the form

$$V(x) = EU(x) + c_0 \cdot \text{Compound} + c_1 \cdot \log|\text{support}(x)|. \quad (16)$$

Note that negative values of c_0 or c_1 indicate complexity aversion; positive values indicate complexity seeking. In robustness checks (available upon request), we model states-aversion using a three-parameter sigmoid function as proposed by Fudenberg and Puri (2021) and find similar results.

Table 10: Parameter estimates for model estimations

Parameter	Model					
	EV	CPT	EV + CA	CPT + CA	EV + CN	CPT + CN
α		0.8 (0.005)		0.78 (0.01)		0.745 (0.006)
β		0.783 (0.01)		0.782 (0.022)		0.74 (0.01)
λ		0.942 (0.048)		0.922 (0.121)		0.879 (0.047)
γ		0.829 (0.004)		0.831 (0.004)		0.81 (0.005)
c_0			-6.295 (0.622)	-0.969 (0.228)		
c_1			0.42 (0.039)	0.43 (0.025)		
η_0	0.128 (0.001)	0.28 (0.006)			0.044 (0.002)	0.153 (0.004)
η_1					0.296 (0.005)	0.609 (0.018)

Notes. Parameter estimates for model estimations described in Section 6. “+ CA” = allows for complexity aversion. “+ CN = allows for complexity-dependent attenuation (η)”.

Estimation results. Table 10 summarizes the parameter estimates across the different models. In our likelihood function, we weight each “person-problem” equally. Though the problems from Peterson et. al. were repeated five times, the repetitions were consecutive, so we do not treat them independently. The choice data we collected contains no repetitions. We report standard errors for the maximum likelihood estimation procedure, which we derive from numerically estimated Hessian matrices.

Alternative specifications. Figure 15 shows the variance explained of a series of alternative models. The first three models are the ones reported in the main paper. The fourth and fifth are analogous except that the reference point is assumed to be the average payout across all menus in each experiment. In the sixth and seventh models, we estimate CARA rather than CRRA utility (with a reference point of zero).

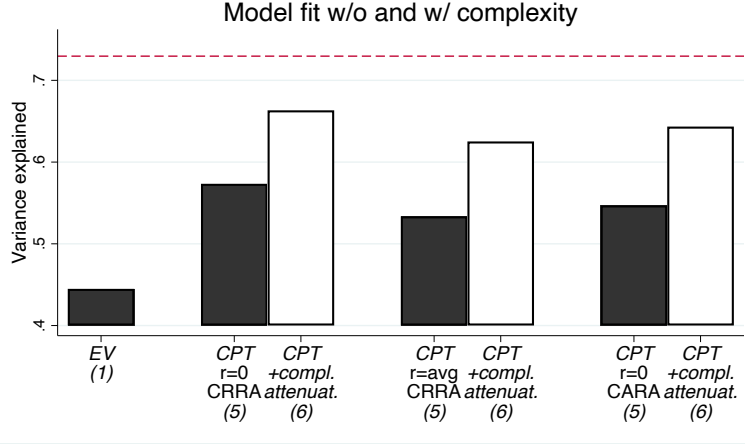


Figure 15: Variance explained of different choice models. Number of estimated model parameters in parentheses. R^2 is computed by first estimating each model, then using estimated model parameters to predict choice rates, and then regressing actual on predicted choice rates in the test set. The first model assumes EV maximization and a constant logit responsiveness term. The second model is the CPT model we estimate throughout the paper, and the third model the extension of CPT in which logit precision depends on complexity. The fourth model is the same CPT model, except that the reference point is assumed to be the average payout across all lotteries in an experiment. The fifth model again augments this CPT model with a complexity-dependent logit precision term. The sixth model is CPT with a reference point of zero and CARA utility, and again the seventh model extends this model with a complexity-dependent logit precision term. For all models, we show the performance in a test set of 2,726 choice problems after the models were estimated in a train sample of 8,172 choice problems. The dashed line corresponds to the performance of a machine learning ensemble.

E Analyses of Within-Subject Experiment

In the *Within Subject* experiment, each participant was confronted with identical problems in a choice frame and an *EV Task* frame. Figure 16 shows the correlation between the elicitations of cognitive uncertainty across these two problem types, controlling for subject fixed effects. These plots are constructed from 300 subjects x 20 problems = 6,000 observations, but we bin them into buckets to make the plot more informative.

Table 11 shows the correlation between cognitive uncertainty in each of the two types of problems with problem features.

Using this data, we can also examine across-subject heterogeneity in the extent to which cognitive uncertainty in the choice and EV tasks are linked. Figure 18 shows the distribution of individual-level correlation coefficients between cognitive uncertainty in choice and in the EV task, where we see substantial heterogeneity. Some of this heterogeneity likely reflects noise or inattention, so we further investigate whether

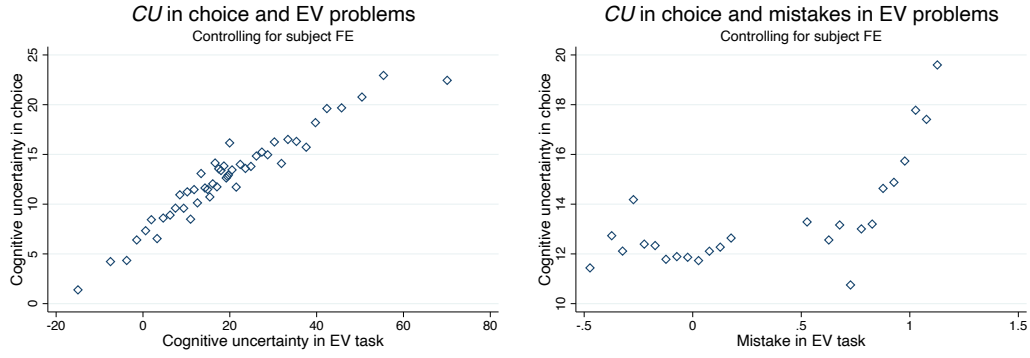


Figure 16: Cognitive uncertainty in choice and in EV task in experiment *Within Subject*. The figures show binned scatter plots of cognitive uncertainty in a choice problem, as a function of cognitive uncertainty in the EV task (left panel) or of a binary mistakes indicator in the EV task. An underlying observation is a subject-decision (not a choice problem). The figure links identical choice sets in the choice and EV task and is constructed controlling for subject fixed effects. Data from 300 subjects and 240 choice sets. Negative values on the x-axis reflect that axes show residuals after netting out subject fixed effects.

Table 11: Correlations of *CU* in choice and *CU* in EV task with complexity features

	CU in choice	CU in EV	Log excess dissim	Log scale	Log support	Frac. lotteries w/ loss	Compound prob
Avg. CU in choice	1.00	0.72	0.64	-0.05	0.25	0.42	0.05
Avg. CU in EV task	0.72	1.00	0.69	0.13	0.25	0.47	0.29
Log excess dissim.	0.64	0.69	1.00	0.40	0.23	0.47	0.00
Ave log scale	-0.05	0.13	0.40	1.00	0.14	-0.12	0.02
Ave log support	0.25	0.25	0.23	0.14	1.00	0.28	-0.14
Frac lotteries w/ loss	0.42	0.47	0.47	-0.12	0.28	1.00	-0.00
Compound prob	0.05	0.29	0.00	0.02	-0.14	-0.00	1.00

Notes. Pairwise Pearson correlations in the *Within Subject* experiment. The first two columns show the correlations of average cognitive uncertainty in choice and the EV task with the complexity features.

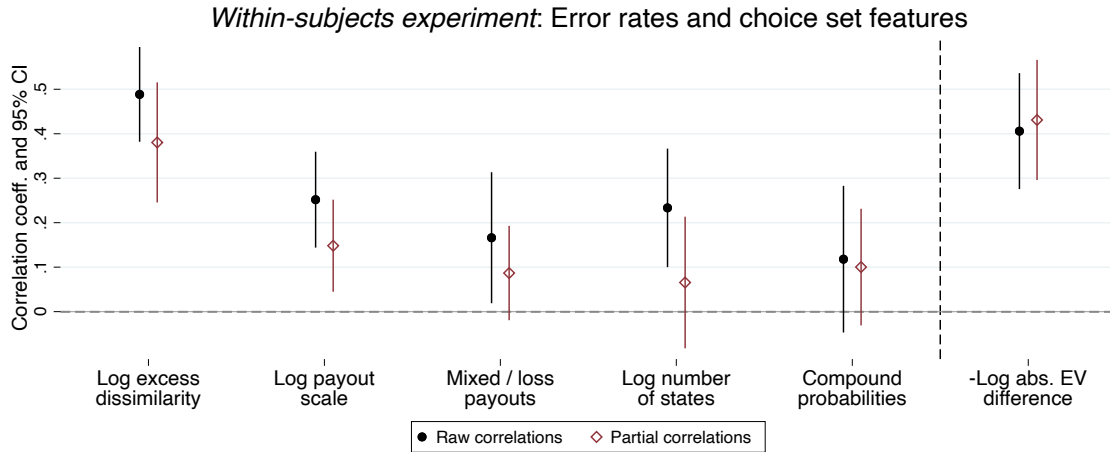


Figure 17: Raw and partial correlation coefficients between task-level error rates and choice set features in the *Within Subject* experiment (240 unique problems). Whiskers show 95% confidence intervals. Partial correlations are calculated controlling for all of the other features in the figure. Log scale, mixed / loss payouts, log number of states and compound probabilities are computed as averages across the lotteries in a set.

correlation strength is linked to choice behavior. We find that across subjects, the CU-correlation is modestly positively correlated ($r = 0.16$, $p = 0.004$) with choosing the EV-maximizing option in the choice task and virtually uncorrelated ($r = 0.05$, $p = 0.42$) with the likelihood of choosing a safe option when available. We view heterogeneity in subjective complexity as an interesting topic for future work.

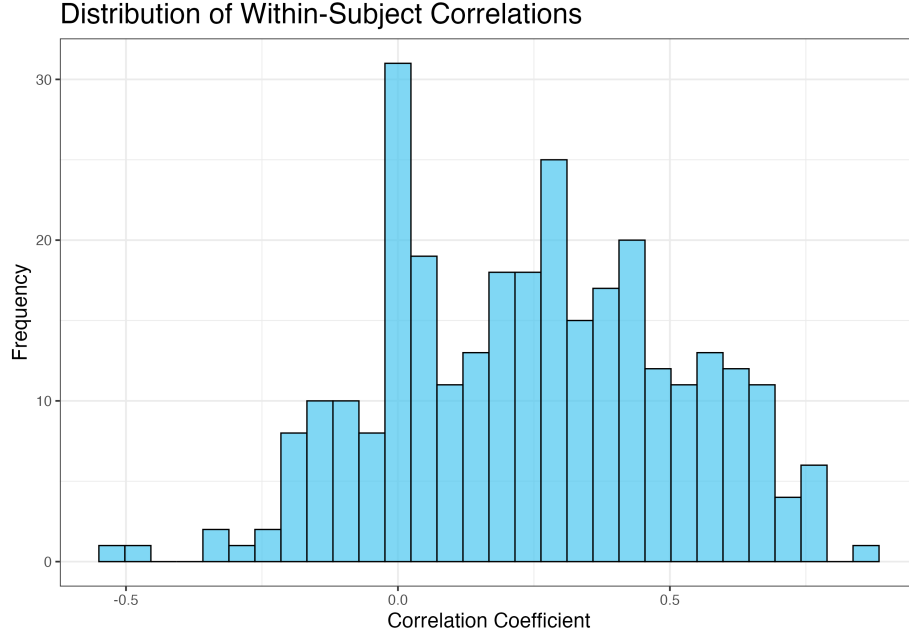


Figure 18: Distribution of individual-level Pearson correlation coefficients between cognitive uncertainty in choice and in the EV task in the *Within Subject* experiment. The mode at 0 is accounted for by 21 subjects who always state the same cognitive uncertainty in at least one of the tasks. The median correlation coefficient is 0.25.

F Extension: Larger Menus

In our *EV Tasks* experiment, we also included 102 menus with between three and five options. These menus were excluded from all analyses in the main paper. Figure 19 shows the link between choice set features and errors / cognitive uncertainty in the full sample of 2,220 problems.

With menus larger than two, calculating excess dissimilarity is less trivial. We proceed by calculating for each pair of options in the set, and then average across all pairs. Similarly, for the lottery-specific features such as payout scale, we average across all lotteries in the set.

Figure 19 shows that menu size has a large effect on both errors and *CU*, though the effect is still somewhat smaller than that of excess dissimilarity.

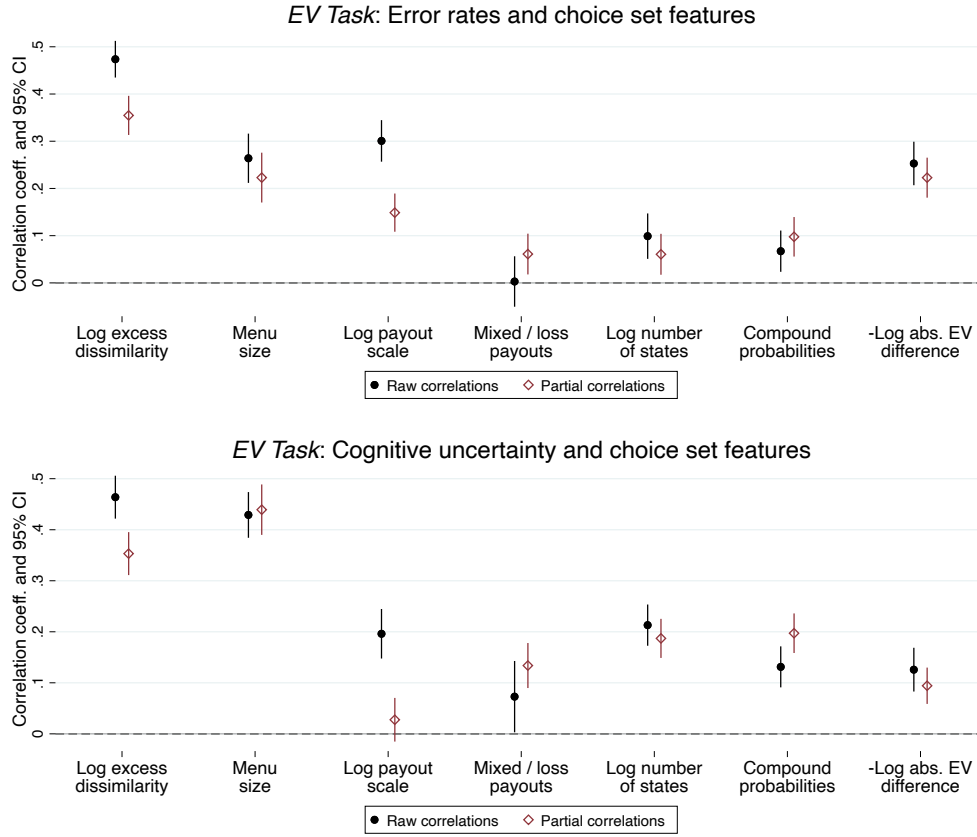


Figure 19: Correlations between choice set features and errors / cognitive uncertainty in the full dataset in the *EV Task*, including menus with more than two options in the train set. Raw and partial correlation coefficients between task-level error rates / average cognitive uncertainty and choice set features. Whiskers show 95% confidence intervals. Excess dissimilarity and proximity of expected values are computed for each pair in the set and then averaged across pairs. Log scale, mixed / loss payouts, log number of states and compound probabilities are computed separately for each lottery and then averaged across the lotteries in a choice set.

G Experimental Instructions and Comprehension Checks

G.1 Experiment *EV Tasks*

Instructions 1/3

Please read these instructions carefully. There will be comprehension checks. If you fail these checks, you will be excluded from the study and you will not receive the completion payment.

In this study, you will make multiple decisions.

Your payment will consist of two components:

- Completion fee:

If you pass all our **comprehension checks and complete the study**, you will receive a **completion fee of \$6**.

- Additional bonus:

On each of 50 decision screens, you will make two decisions. One of the decision screens will be selected at random by the computer, with equal probability. The computer will then flip a coin to determine which of your two decisions on that screen will determine your **bonus**. The **maximal bonus** you can earn in this study is **\$10**. Your bonus will actually be paid out to you with probability 1/2.

Instructions 2/3

Guessing task: which lottery has the highest average payout?

On each decision screen, you will be presented with multiple lotteries that pay out different amounts of money with different probabilities.

Your task is NOT to indicate which lottery you would personally prefer to receive.

Instead, you will be **asked to guess which lottery has the highest average payout**, in the following sense:

- Each lottery is run many, many times (100,000 times). In each run, we **record the payout** of the lottery. In the end, **for each lottery**, we **compute the average payout** across all runs.
 - For some lotteries, you will not know the precise probabilities. Instead, we will tell you how the probabilities are determined.
- On each decision screen, you will make two decisions that reflect which lottery you think has the highest average payout.

Step 1: Guess the lottery with the highest average payout

- We will ask you to indicate which lottery you think produces the highest average payout. You need to **select exactly one lottery**.
- This question has a mathematically correct solution.
- If this decision is randomly chosen for payment, you will receive \$10 if your answer is correct, and nothing otherwise.

Step 2: Indicate your certainty about your guess

- We will ask you to **allocate 100 “certainty points” between the different lotteries** to indicate how likely you think it is that each lottery has the highest average payout.
 - If you are uncertain, you should allocate some points to each lottery you think could have the highest average payout.
 - For example, **if you think it is 80% likely that lottery A has the highest average payout, you should allocate 80 certainty points to Lottery A.**
 - If you are certain which lottery has the highest average payout, you should allocate all 100 points to that lottery.
- If this decision is randomly chosen for payment, you will receive 10 cents for each certainty point that you allocate to the lottery that actually delivers the highest average payout. Certainty points allocated to incorrect lotteries earn nothing.

Example screen:

Which lottery has the highest average payout if the computer runs it many, many times?
Please select one.

Lottery A	Lottery B
Prob. 50%: Get \$30 Prob. 50%: Get \$0	Prob. 100%: Get \$9

How certain are you that each lottery has the highest average payout?
Please allocate 100 certainty points.

points

points

100 points left to allocate.

In this example, if Lottery A was run many, many times, it would have a higher average payout (\$15) than lottery B if it was run many, many times (\$9).

Here is how your bonus would be determined in this example:

- If Step 1 was selected for payment, you'd get a \$10 bonus if you indicated Lottery A, and nothing otherwise.
- If Step 2 was selected for payment, you'd get 10 cents for each certainty point that you allocated to Lottery A. For example, if you had allocated 80 points to A and 20 points to B, your bonus would be \$8.

Instructions 3/3

In total, you will complete 50 decision tasks. You may take as much time for each task as you'd like, though remember that the study was advertised for 35 minutes and you will only be paid on that basis. If you find that you don't have much time, you may look at the lotteries and make an informed guess about which one has the highest average payout. But again, it is up to you how you work on the tasks.

Once you click the next button, the comprehension check questions will start!

Comprehension check

To verify your understanding of the instructions, please answer the comprehension questions below. If you get one or more of them wrong twice in a row, you will not be allowed to participate in the study and earn a completion payment. In each question, exactly one response option is correct.

You can review the instructions [here](#).

1. Which of the following statements is correct?

One of my tasks is to indicate which lottery I would personally prefer to receive.

One of my tasks is to guess which lottery has the highest average payout if the computer runs it many, many times.

2. On each decision screen, you are asked to indicate which lottery has the highest average payout. How is your potential bonus for this part determined?

My bonus will be \$10 if I indicated the lottery with the highest average payout and \$0 otherwise.

My bonus depends on the outcome of the lottery that I selected. Thus, if the lottery has negative values, it's possible for me to make a loss.

3. Which of the following statements is correct?

On each decision screen I will be asked to allocate 100 certainty points to indicate how certain I am about which lottery I would personally prefer to receive.

On each decision screen I will be asked to allocate 100 certainty points to indicate how likely I think it is that a lottery has the highest average payout.

4. Please select the statement that is true about the certainty points.

If I think it is 60% likely that Lottery A has the highest average payout, I should allocate 60 certainty points to A.

I should always allocate all certainty points to one lottery, even if I'm not certain which one has the highest average payout.

I should always allocate some certainty points to each lottery, even if I'm certain which one has the highest average payout.

If I think it is 60% likely that Lottery A has the highest average payout, I should allocate 100 certainty points to A.

G.2 Experiment *Choice Tasks*

Instructions 1/2

Please read these instructions carefully. There will be comprehension checks. If you fail these checks, you will be excluded from the study and you will not receive the completion payment.

In this study, you will make multiple decisions.

Your payment will consist of two components:

- Completion fee:

If you pass all our **comprehension checks and complete the study**, you will receive a **completion fee of \$3.50**.

- Additional bonus:

On each of 50 decision screens, you will make a decision. One of the decision screens will be selected at random by the computer, with equal probability, and will determine your **bonus**. The **maximal bonus** you can earn in this study is **\$320**. Your bonus will actually be paid out to you with probability 1/5.

Instructions 2/2

Choice task: which lottery would you like to receive?

On each decision screen, you will be presented with two lotteries that pay out different amounts of money with different probabilities. The computer actually plays out these lotteries according to the probabilities we specify, and pays you accordingly.

On each decision screen, you will be asked to indicate which lottery you prefer to receive.

Step 1: Choose the lottery you prefer

- We will ask you to indicate which lottery you prefer to receive.
- If this decision is randomly chosen for payment, the computer will play out the selected lottery. The outcome of this lottery will determine your potential bonus.

Step 2: Indicate your certainty about your choice

- You might feel uncertain about which lottery you actually prefer. Therefore, we will ask you to indicate how certain you are (in percent) that you actually prefer the lottery that you chose.
 - For example, if you think it is 70% likely that you actually prefer the lottery that you chose, you should set the slider to 70%.
 - If you are certain that you prefer the lottery you chose, you should set the slider to 100%.

For each choice problem in which you can potentially incur a loss, you will receive a budget. This budget will always be given by the largest amount of money you can lose in that choice problem. The budget cannot be transferred across decisions. If a decision of yours is selected to determine your bonus and it happens to be a loss, then this loss will be subtracted from the budget in that particular task, thus giving your final bonus. If you don't incur a loss, the budget will simply be added to your payout from the lottery.

For some lotteries, you will not know the precise probabilities. Instead, we will tell you how the probabilities are determined.

Example screen:

Which lottery do you choose?
Please select one.

Lottery A

Probability 50%: Get \$30
Probability 50%: Get \$0

Lottery B

Probability 100%: Get \$9

How certain are you that you actually prefer the lottery you chose above?

Fully certain I prefer the lottery I didn't choose

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

Fully certain I prefer the lottery I chose

I am **PLEASE CLICK SLIDER** certain that I actually prefer the lottery I chose above.

Here is how your bonus would be determined in this example:

- If you selected Lottery A, the computer would actually play out Lottery A, so your bonus would be \$30 with 50% probability and nothing with 50% probability.
- If you selected Lottery B, your bonus would be \$9.

After your bonus is determined, the computer will randomly determine whether or not it will be paid out to you. You will actually receive your bonus 1/5 of the time.

Once you click the next button, the comprehension check questions will start!

Comprehension check

To verify your understanding of the instructions, please answer the comprehension questions below. If you get one or more of them wrong twice in a row, you will not be allowed to participate in the study and earn a completion payment. In each question, exactly one response option is correct.

You can review the instructions [here](#).

1. How is your bonus determined?

I will make multiple decisions, and every one of them will get paid. Thus, I can strategize across decisions.

I will make multiple decisions. The computer will randomly select one of them, and my potential bonus will depend on my decision in this one question. Thus, there is no point for me in strategizing across decisions.

2. Suppose that you picked Lottery A in one of the tasks.

Lottery A

Probability 60%: **Get \$20**

Probability 40%: **Get \$3**

Which of the following statements is correct?

I know that my payoff from this lottery will be either a profit of \$20 OR a profit of \$3, but not both.

I know that my payoff from this lottery will be a profit of \$20 AND a profit of \$3.

3. Which of the following statements is correct?

On each decision screen I will be asked to indicate how certain I am that my bonus will be determined by each lottery.

On each decision screen I will be asked to indicate how certain I am that I actually prefer the lottery that I chose.

4. Please select the statement that is true about the certainty slider.

If I think it is 60% likely that I actually prefer the lottery that I chose, I should set the certainty slider to 60%.

If I think it is 60% likely that I actually prefer the lottery that I chose, I should nonetheless set the certainty slider to 100% because I chose a specific lottery.

G.3 Experiment *Within Subject*

The order of the two tasks in our *Within Subject* experiment was randomized. The experiment was structured so that participants first saw a set of general instructions, followed by the instructions for the first task. After completing the first task along with comprehension checks, participants then saw instructions for the second task and associated comprehension checks. The task instructions are virtually independent of order, except that two of the comprehension questions apply to both tasks, and so are only asked for the first. The instructions below have *Choice* tasks first.

General Instructions

Please read these instructions carefully. There will be comprehension checks. If you fail these checks, you will be excluded from the study and you will not receive the completion payment.

There are two parts to this study. In both parts, you will make multiple decisions.

Your payment will consist of two components:

- **Completion fee:** If you pass all our **comprehension checks and complete both parts of the study**, you will receive a **completion fee of \$3.50**.
- **Additional bonus:** In each part of the study, you will face 20 decision screens where you will make a decision. One of these decision screens will be selected at random by the computer, with equal probability, and will determine your **bonus**. Your bonus will actually be paid out to you with probability 1/5.

Budget

- For some of the decisions which are eligible for a bonus payment, it is possible to make a "loss." If such a decision is selected to determine your bonus, you will receive a budget.
- This budget will always be equal to the largest loss you can potentially incur for that decision.
 - If your decision happens to earn a loss, then this loss will be subtracted from the budget in that particular task to give your final bonus.
 - If you don't incur a loss, the budget will simply be added to whatever you earn from your decision.
- The budget cannot be transferred across decisions.

Part 1 Instructions

Choice task: which lottery would you like to receive?

On each decision screen, you will be presented with two lotteries that pay out different amounts of money with different probabilities. The computer actually plays out these lotteries according to the probabilities we specify, and pays you accordingly.

On **each decision screen**, you will be asked to indicate which lottery you prefer to receive.

Step 1: Choose the lottery you prefer

- We will ask you to indicate which lottery you prefer to receive.
- If this decision is randomly chosen for payment, the computer will play out the selected lottery. The outcome of this lottery will determine your potential bonus.

Step 2: Indicate your certainty about your choice

- You might feel uncertain about which lottery you actually prefer. Therefore, we will ask you to indicate how certain you are (in percent) that you actually prefer the lottery that you chose.
 - For example, if you think it is 80% likely that you actually prefer the lottery that you chose, you should set the slider to 80%.
 - If you are certain that you prefer the lottery you chose, you should set the slider to 100%.

Unknown Probabilities

For some lotteries, you will not know the precise probabilities. Instead, the probabilities will be randomly drawn, and we will explain the random process in the instructions for that choice.

Example screen:

The screenshot shows a decision screen titled "Which lottery do you choose? Please select one." It presents two options: Lottery A and Lottery B. Lottery A has two outcomes: "Probability 50%: Get \$30" and "Probability 50%: Get \$0". Lottery B has one outcome: "Probability 100%: Get \$9". Below the lotteries, a question asks, "How certain are you that you actually prefer the lottery you chose above?". A horizontal slider is provided, ranging from 0% to 100% in 10% increments. The left end is labeled "Fully certain I prefer the lottery I didn't choose" and the right end is labeled "Fully certain I prefer the lottery I chose". A vertical line on the slider indicates the current certainty level. At the bottom, a text prompt says, "I am PLEASE CLICK SLIDER certain that I actually prefer the lottery I chose above."

Here is how your bonus would be determined in this example:

- If you selected Lottery A, the computer would actually play out Lottery A, so your bonus would be \$30 with 50% probability and nothing with 50% probability.
- If you selected Lottery B, your bonus would be \$9.

After your bonus is determined, the computer will randomly determine whether or not it will be paid out to you. You will actually receive your bonus 1/5 of the time.

Once you click the next button, the comprehension check questions will start!

Comprehension Check (Part 1)

To verify your understanding of the instructions, please answer the comprehension questions below. If you get one or more of them wrong twice in a row, you will not be allowed to participate in the study and earn a completion payment. In each question, exactly one response option is correct.

You can review the instructions [here](#).

1. How is your bonus determined?

I will make multiple decisions. The computer will randomly select one of them, and my potential bonus will depend on my decision in this one question. Thus, there is no point for me in strategizing across decisions.

I will make multiple decisions, and every one of them will get paid. Thus, I can strategize across decisions.

2. Suppose that you picked Lottery A in one of the tasks.

Lottery A

Probability 60%: Get \$20
Probability 40%: Get \$3

Which of the following statements is correct?

I know that my payoff from this lottery will be either a profit of \$20 OR a profit of \$3, but not both.

I know that my payoff from this lottery will be a profit of \$20 AND a profit of \$3.

3. Which of the following statements is correct?

On each decision screen I will be asked to indicate how certain I am that I actually prefer the lottery that I chose.

On each decision screen I will be asked to indicate how certain I am that my bonus will be determined by each lottery.

4. Please select the statement that is true about the certainty slider.

If I think it is 60% likely that I actually prefer the lottery that I chose, I should set the certainty slider to 60%.

If I think it is 60% likely that I actually prefer the lottery that I chose, I should nonetheless set the certainty slider to 100% because I chose a specific lottery.



Part 2 Instructions

Guessing task: which lottery has the highest average payout?

On each decision screen in the second part of the study, you will again be presented with two lotteries that pay out different amounts of money with different probabilities.

This time, your task is **NOT** to indicate which lottery you would personally prefer to receive if it was only played out once.

Instead, you will be paid according to the average payout of each lottery, in the following sense:

- **Each lottery is run many, many times** (100,000 times). In each run, we **record the payout** of the lottery. In the end, for each lottery, we **compute the average payout across all runs**.
 - We know from statistics that when a lottery is run so many times, we can trust that the probabilities will match the true shares of payouts. Put simply, after running a lottery that pays \$4 with probability 30% many, many times, statistics tells us that 30% of the payouts will *actually* be \$4.
- **On each decision screen, you will make two decisions that reflect which lottery you think has the highest average payout.**

Step 1: Select lottery with highest average payout

- We will ask you to indicate which lottery you think produces the highest average payout. You need to select exactly one lottery.
- This question has a mathematically correct solution.
- If this decision is randomly chosen for payment, you will receive **the average payout of the lottery you selected**.
- Your payment is based only on the **average payout** of the lotteries. This means it is always in your best interest to choose the lottery which pays out more on average, even if that lottery is riskier.

Step 2: Indicate your certainty about your choice

- You might feel uncertain about which lottery has the highest average payout. Like in the previous part, we will ask you to indicate how certain you are (in percent) that you actually chose the lottery with the highest average payout.

Unknown Probabilities

For some lotteries, you will not know the precise probabilities. Instead, the probabilities will be randomly drawn, and we will explain the random process in the instructions for that choice. To determine how much each lottery pays on average, the probabilities are randomly drawn many, many times. Each time, the lottery is run with the chosen probabilities and the payout is recorded. We then compute the average payout across all of these draws.

Example screen:

Which lottery has the highest average payout if the computer ran it many, many times?
Please select one.

Lottery A	Lottery B
Probability 60%: Get \$6 Probability 40%: Get \$1	Probability 100%: Get \$7

How certain are you that you actually selected the lottery with the highest average payout?

Fully certain I did not choose the lottery with highest average payout | Fully certain I chose the lottery with highest average payout

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

I am **PLEASE CLICK SLIDER** certain that I selected the lottery with the highest average payout.

In this example, if Lottery A was run many, many times, it would have a lower average payout (\$5.20) than Lottery B if it was run many, many times (\$7).

Here is how your bonus would be determined in this example:

- If you selected lottery A, your bonus would be \$5.20
- If you selected lottery B, your bonus would be \$7.

After your bonus is determined, the computer will randomly determine whether or not it will be paid out to you. You will actually receive your bonus 1/5 of the time.

Once you click the next button, the comprehension check questions will start!

Comprehension Check (Part 2)

To verify your understanding of the instructions, please answer the comprehension questions below. You must answer each question correctly to participate in the study and earn a completion payment. In each question, exactly one response option is correct.

You can review the instructions [here](#).

1. Suppose that you picked Lottery B in one of the tasks and this decision is selected to earn a bonus payment.

Lottery B

Probability 85%: **Get \$9**
Probability 15%: **Get \$4**

Which of the following statements is correct?

I know that my payoff from this lottery will be a profit of \$9 AND a profit of \$4, for a total of \$13.

I know that my payoff from this lottery will be either a profit of \$9 OR a profit of \$4, but not both.

I know that my payoff from this lottery will be equal to the average payoff of the lottery, which is \$8.25.

2. Which of the following statements is correct?

On each decision screen I will be asked to indicate how certain I am that I selected the lottery with the highest average payout.

On each decision screen I will be asked to indicate how certain I am that I actually prefer the lottery that I selected.

On each decision screen I will be asked to indicate how certain I am that my bonus will be determined by each lottery.